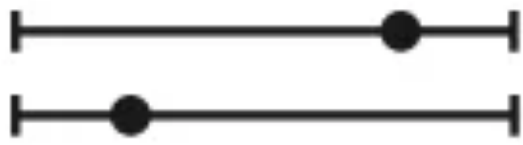
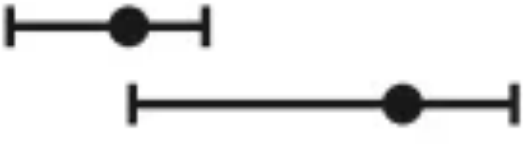


➔ **Magnitude Channels: Ordered Attributes**

Position on common scale



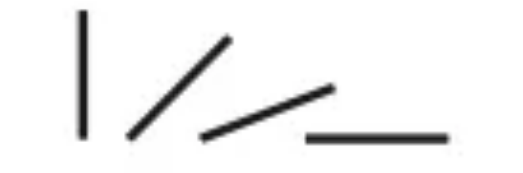
Position on unaligned scale



Length (1D size)



Tilt/angle



Area (2D size)



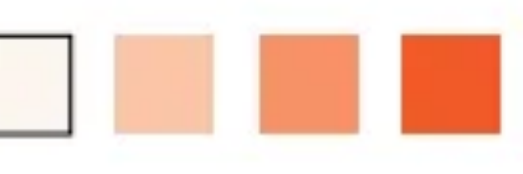
Depth (3D position)



Color luminance



Color saturation



Curvature



Volume (3D size)

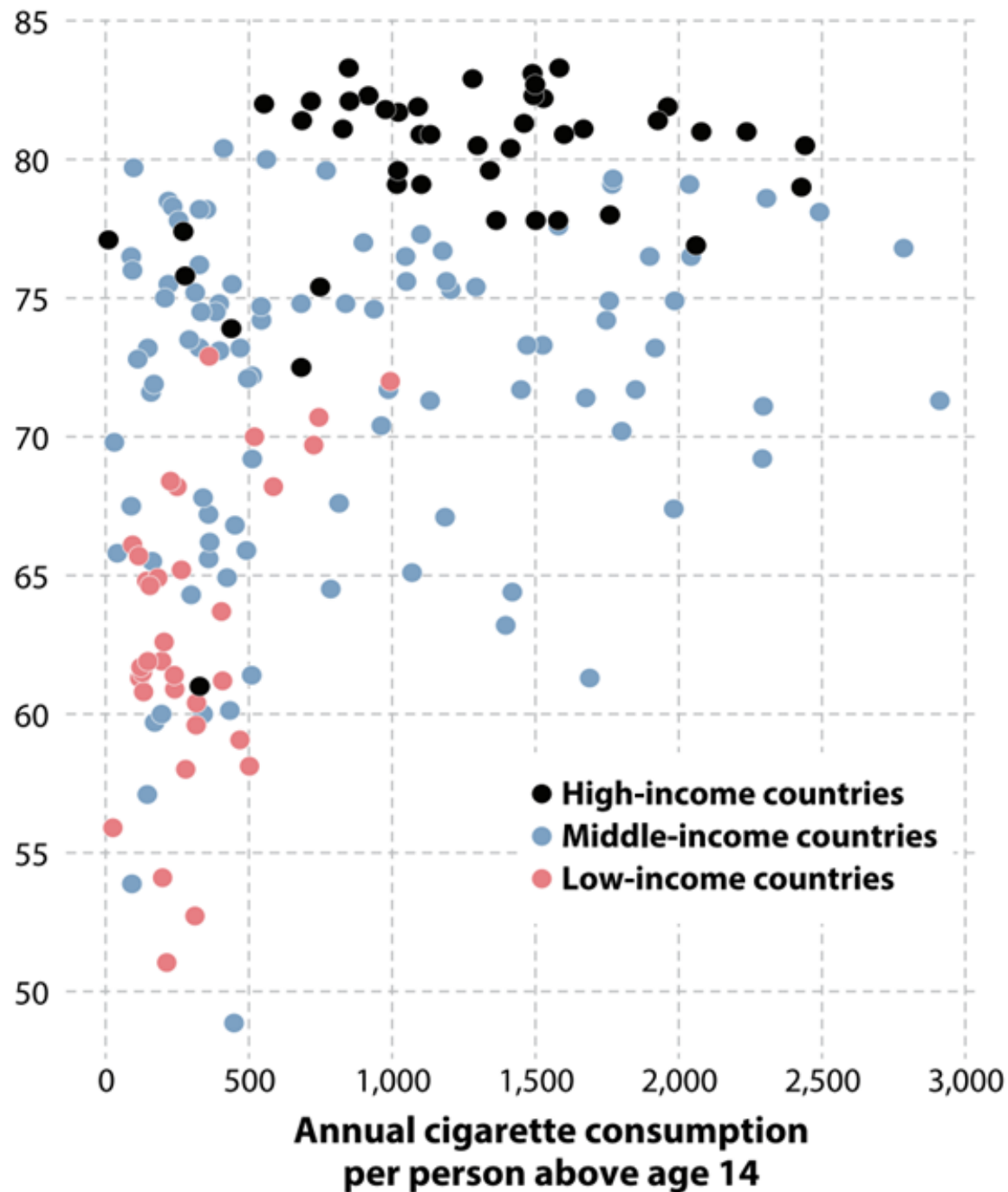
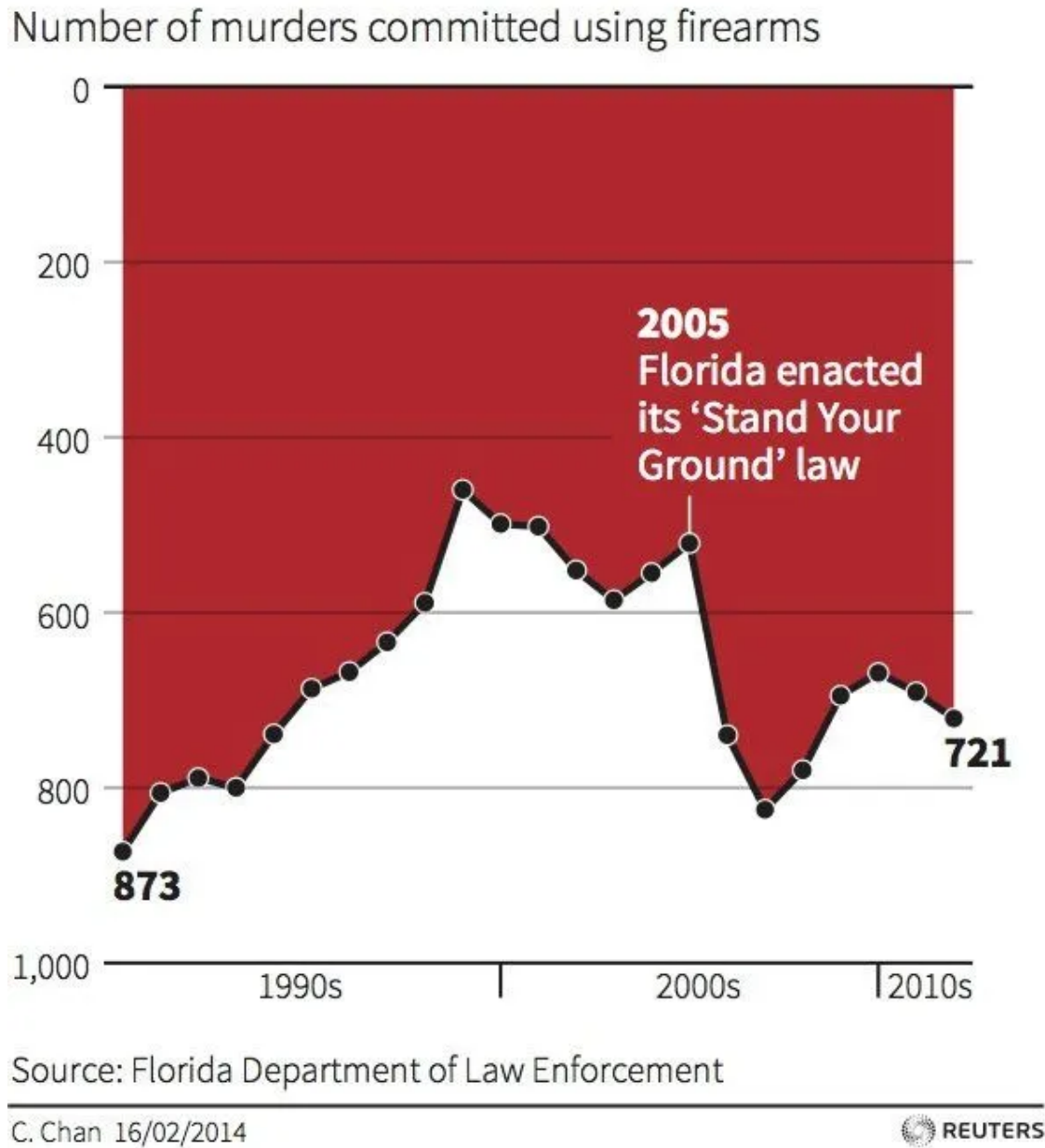


Data

Expressiveness

Visual Effectiveness

Gun deaths in Florida



Week 3: Perception and Cognition

CMSC471: Introduction to Information Visualization

Fumeng Yang, Assistant Professor

February 16, 2026

University of Maryland, College Park

<https://fumeng-yang.github.io/CMSC471>

Human **perception** and **cognition** can help
us go further than just channel selection

Perceptual processes

Perception and cognition

Signal detection

Magnitude estimation

Pre-attentive processing

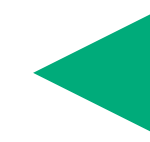
Selective attention

Change blindness

Gestalt grouping

Perception and cognition

Signal detection



Magnitude estimation

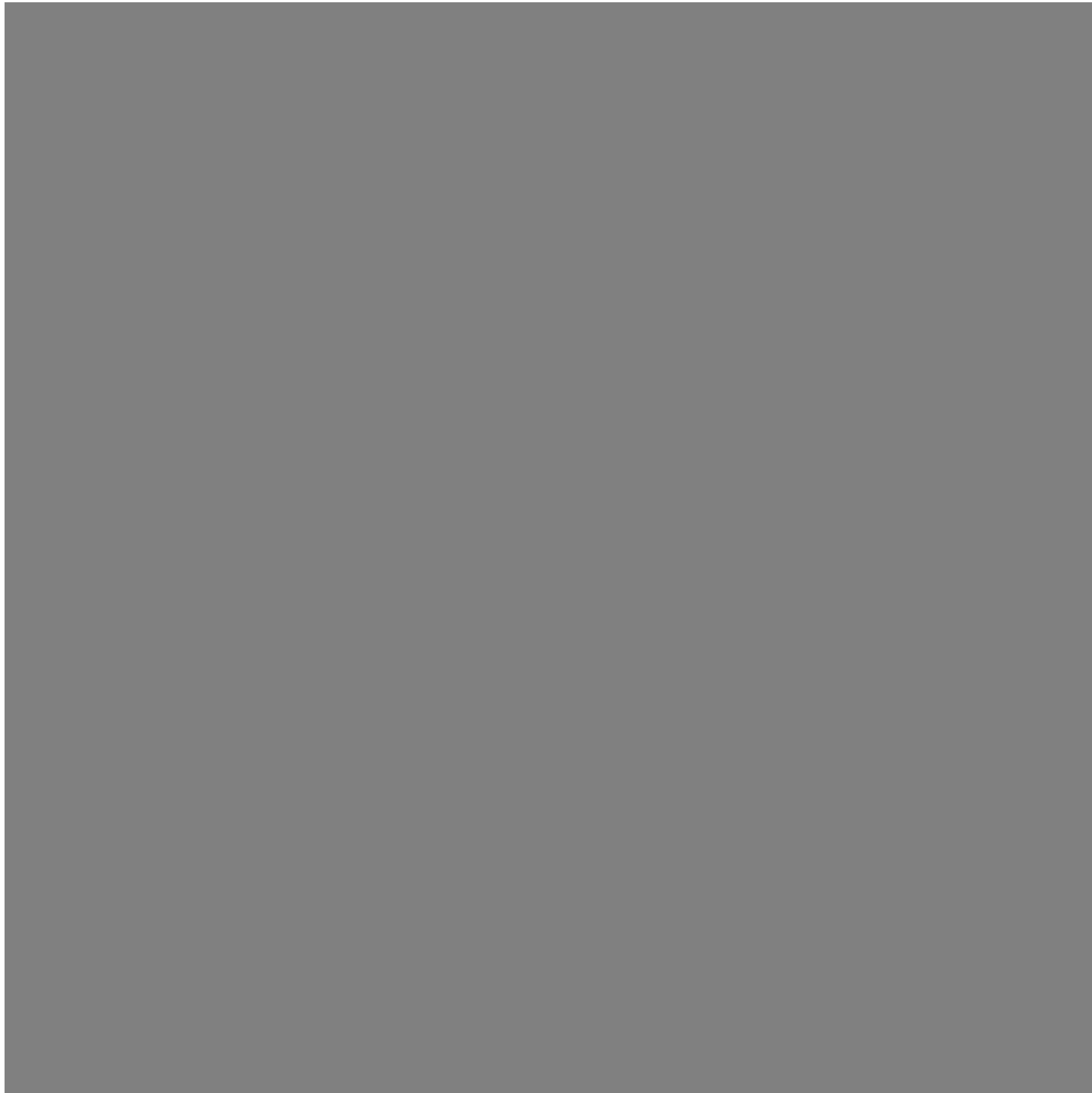
Pre-attentive processing

Selective attention

Change blindness

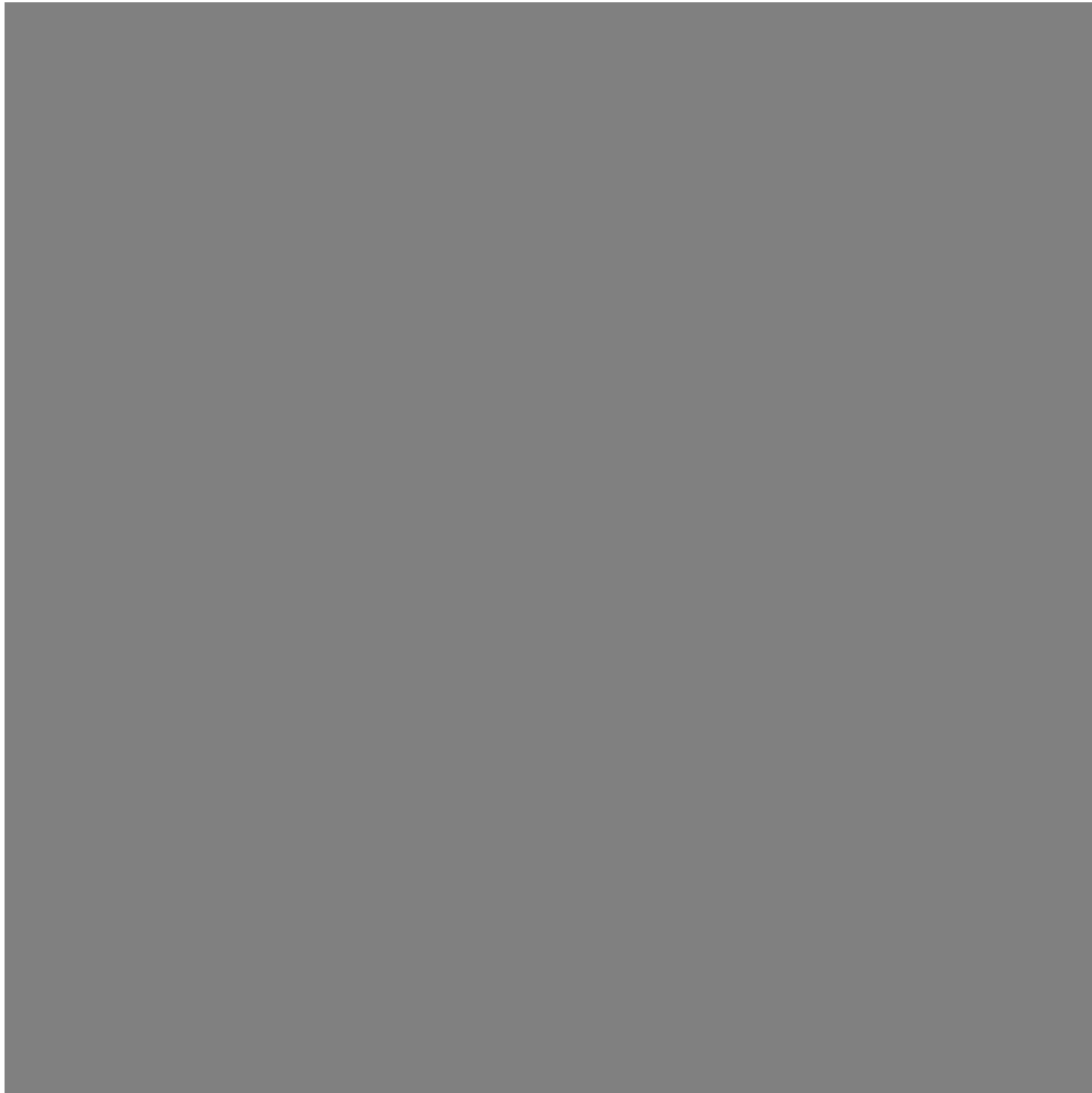
Gestalt grouping

Which is brighter?



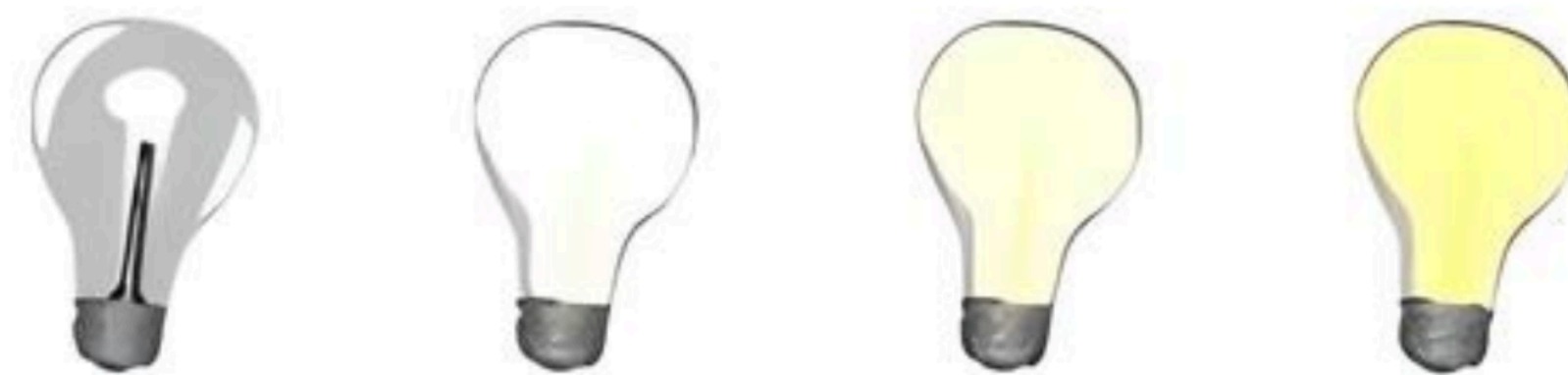
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Which is brighter?

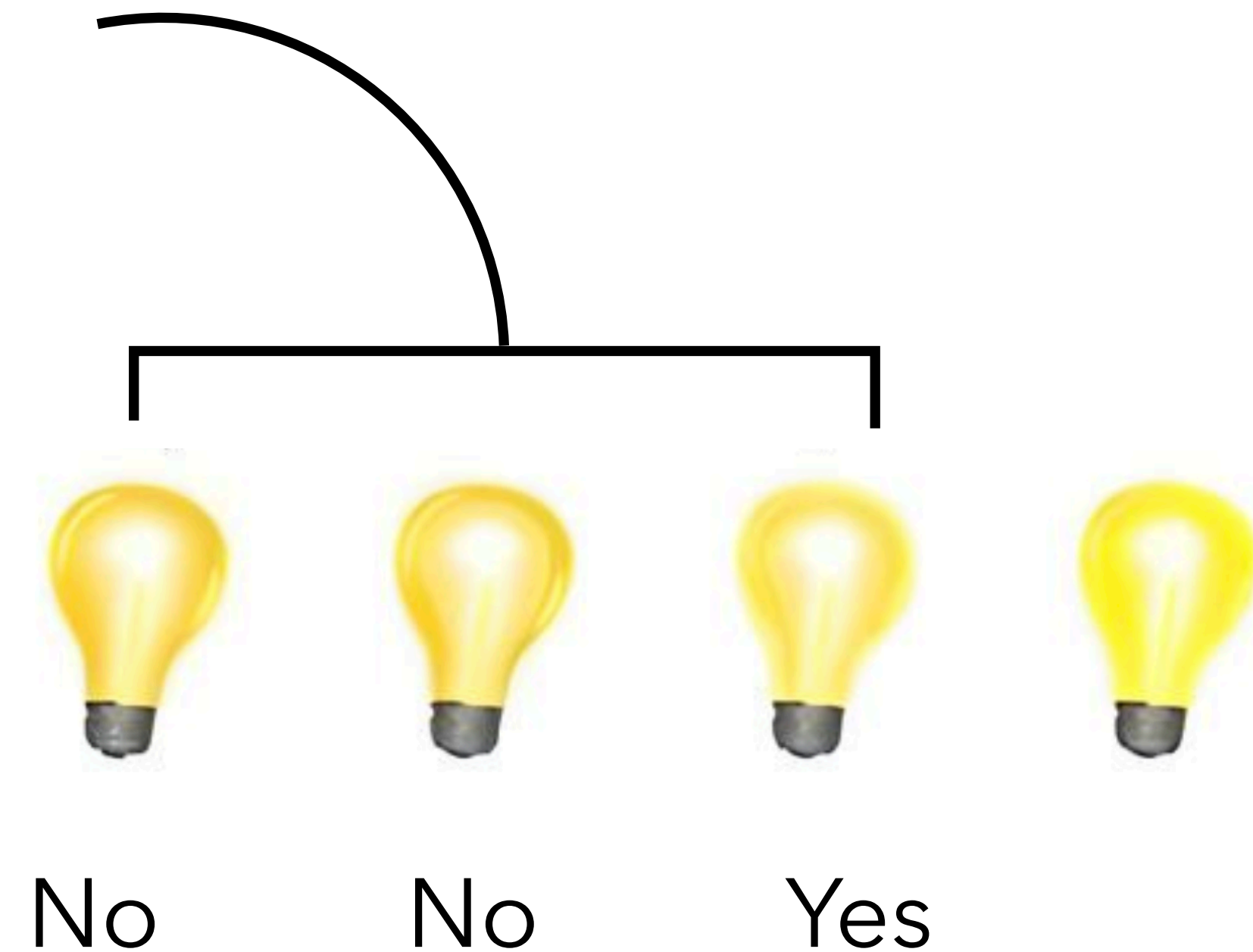


Smallest difference we can tell

The minimum threshold between two stimuli required for detecting 50% (or 75%) of the time, called just noticeable difference (JND)



Area, length, brightness, pressure,
cold, pain, warmth...



Observer's Responses

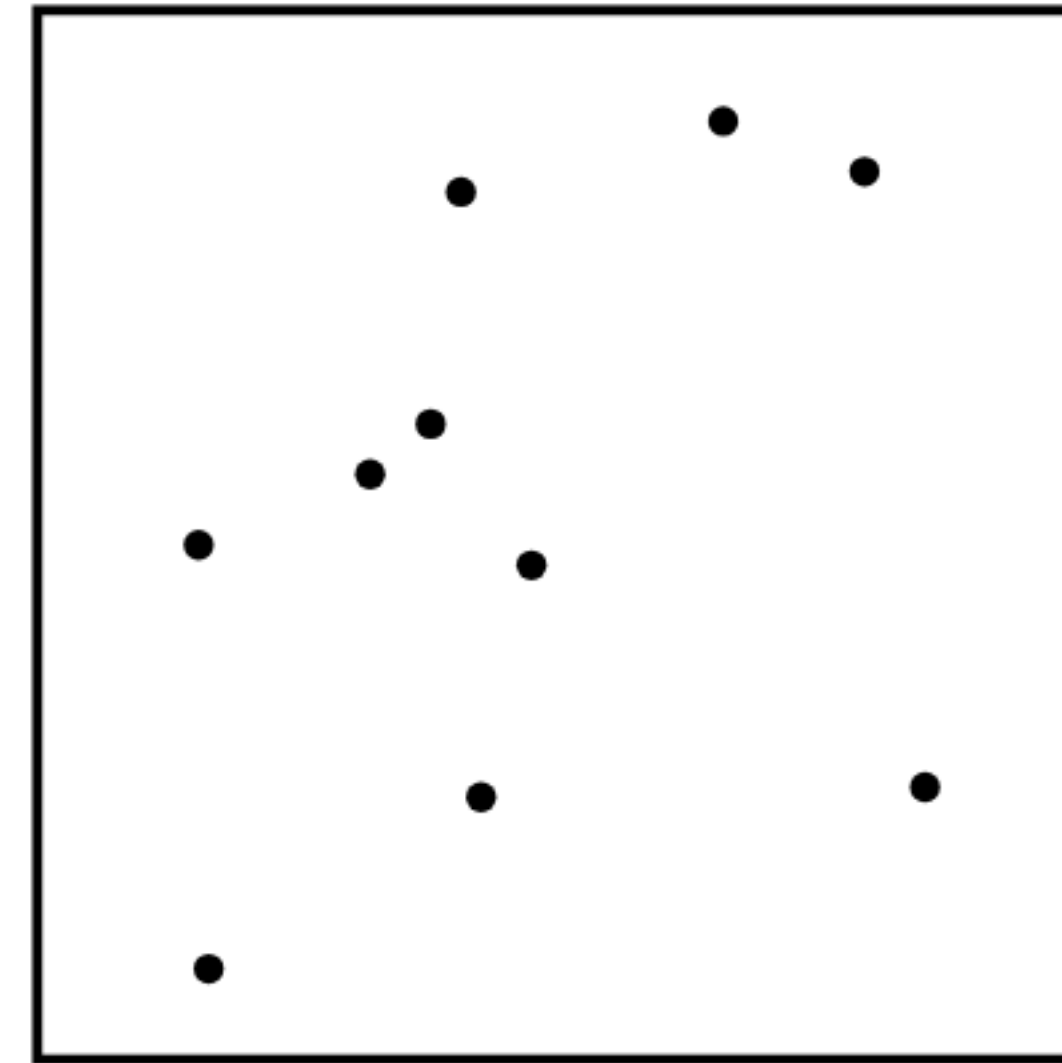
Weber's law

$$\Delta I = kI$$

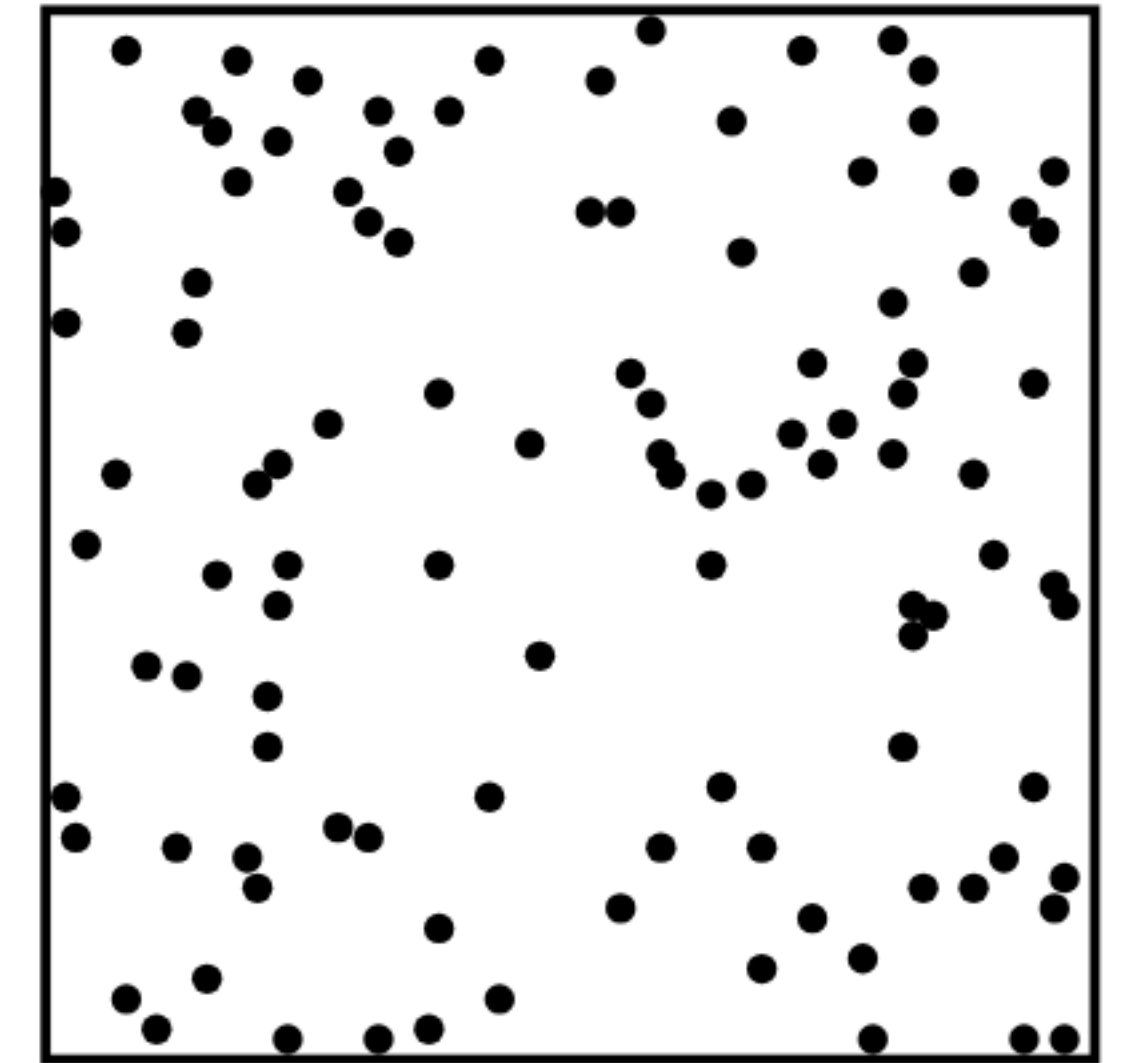
JND as a function of magnitude/
intensity.



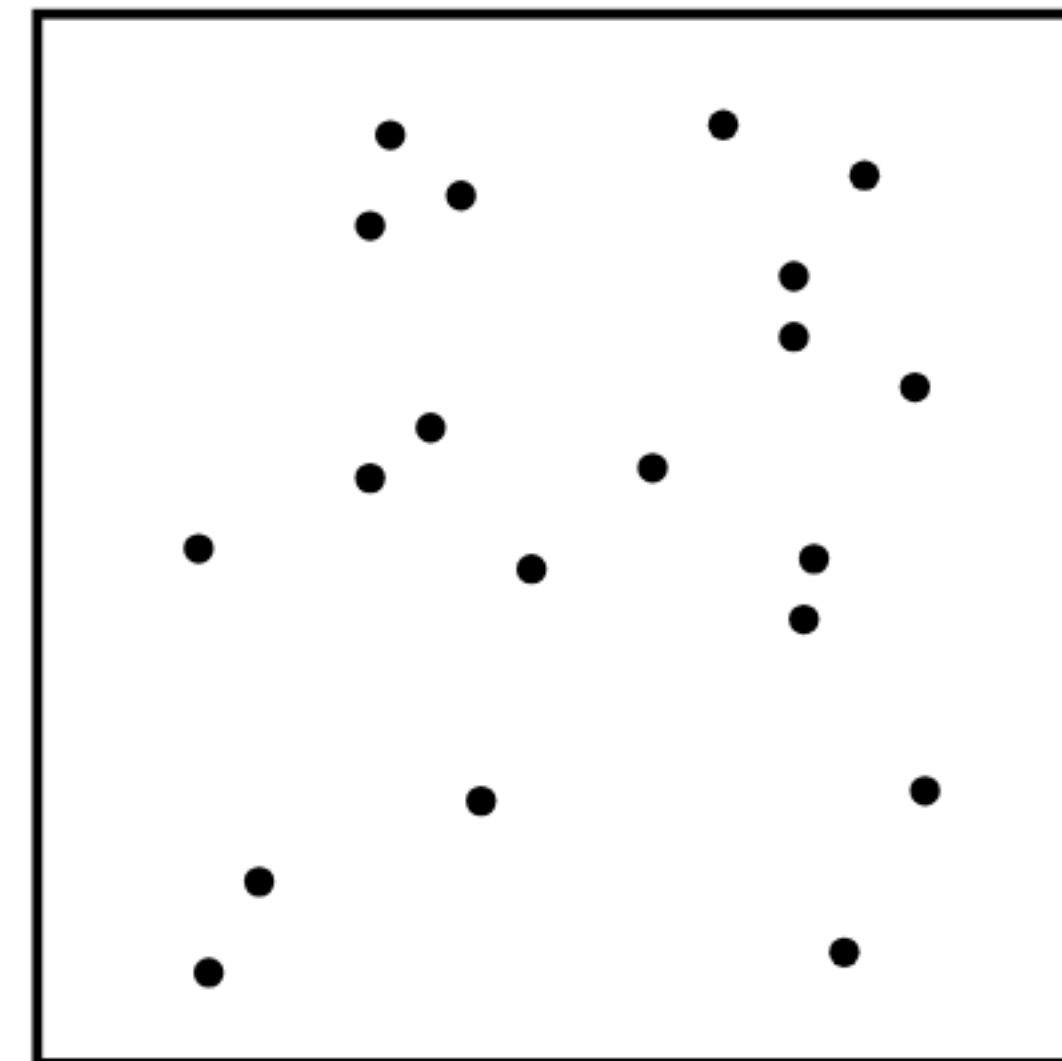
Ernst Heinrich Weber
German physiologist
1795 - 1878



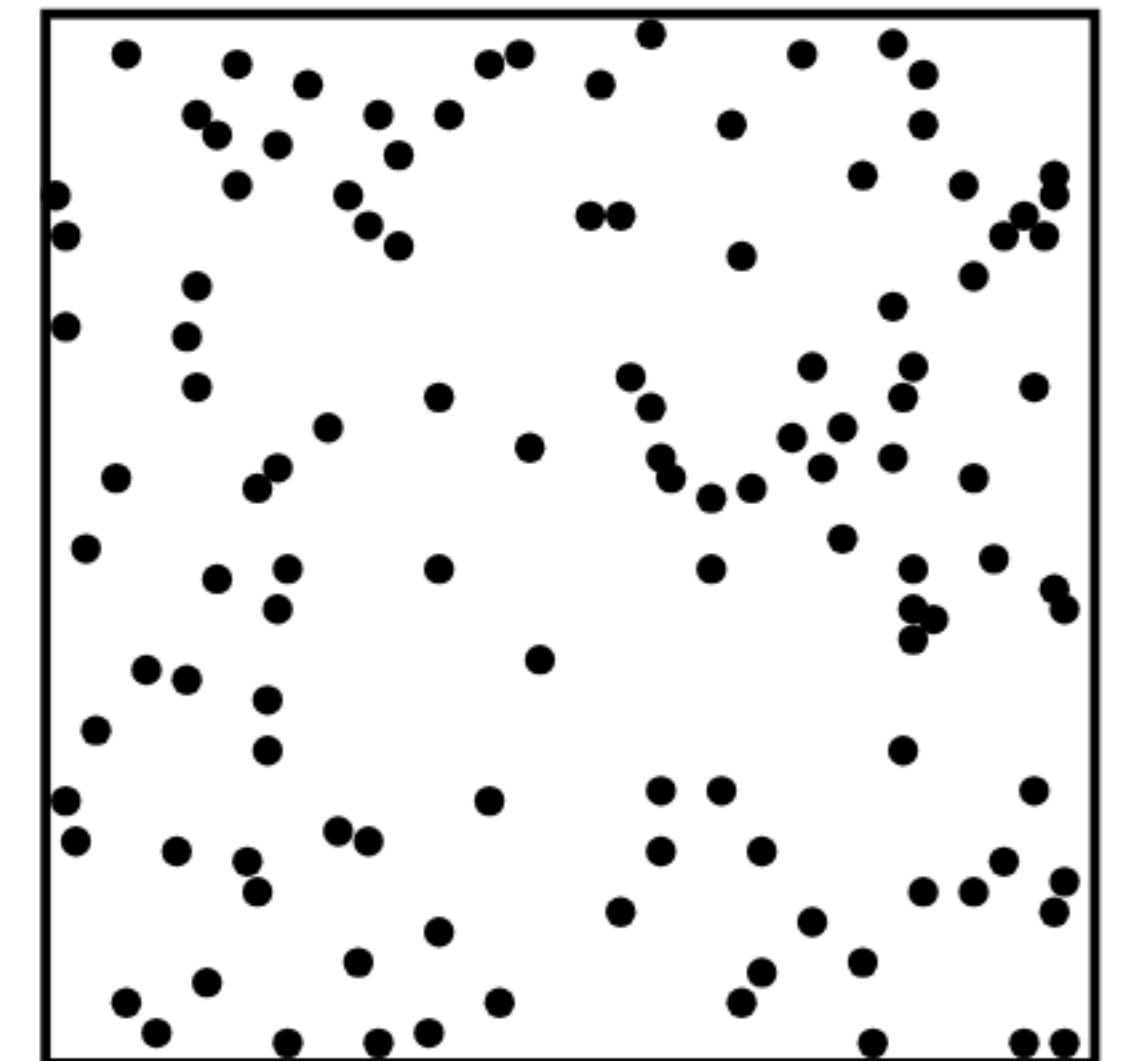
10



110

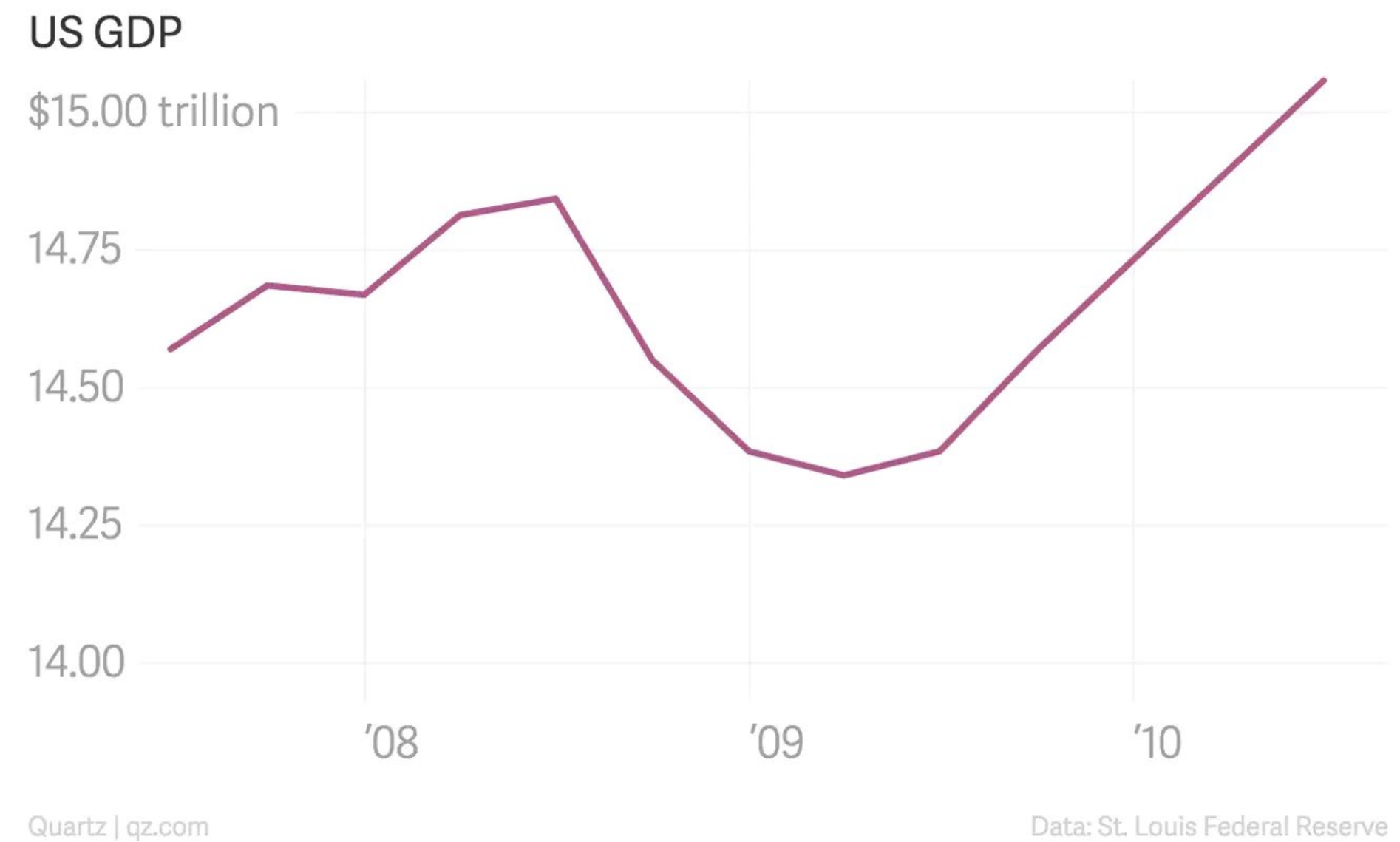
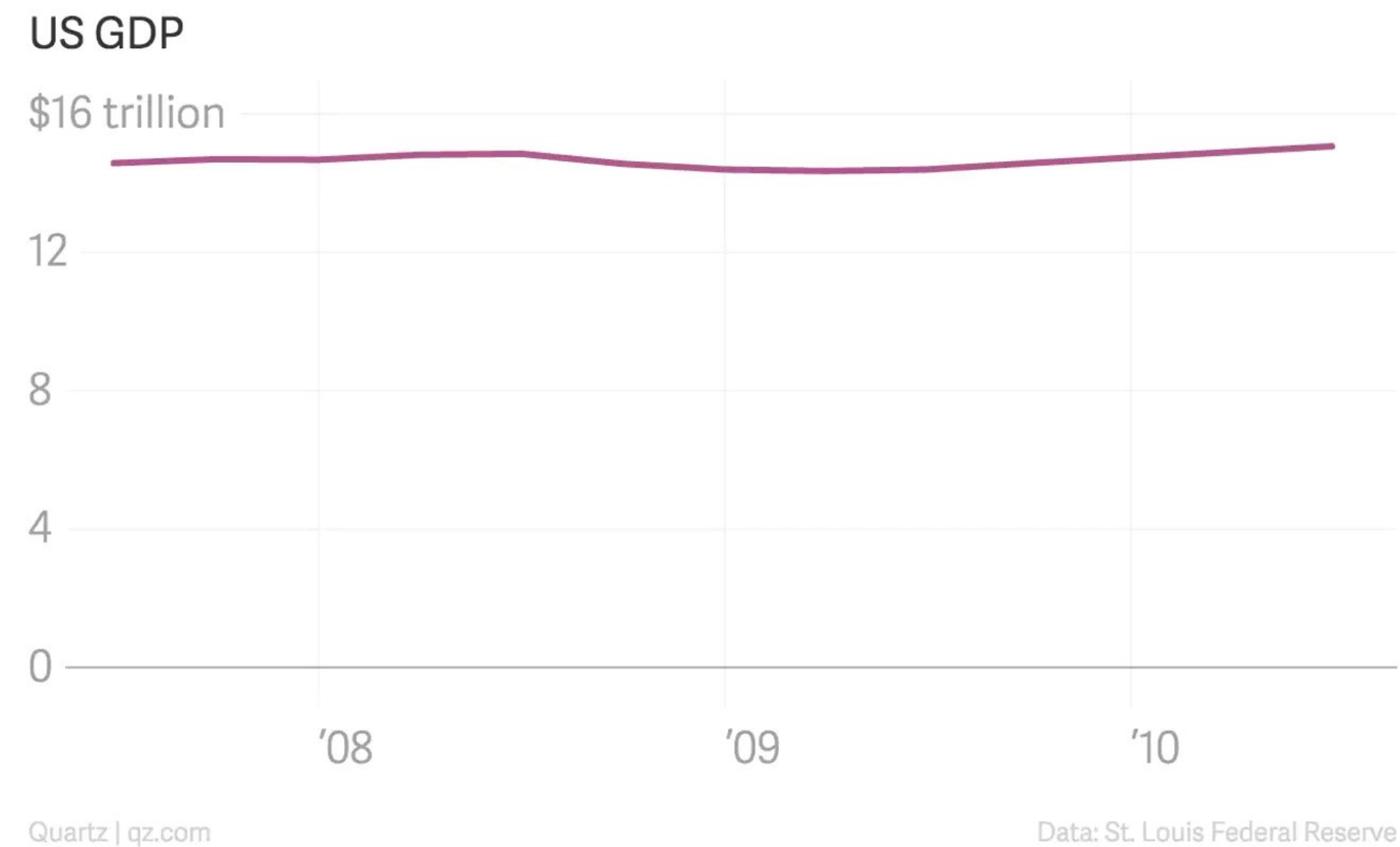


20



120

Truncate the y-axis: lie? effective?



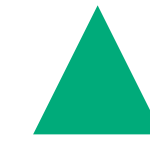
If the zero value doesn't make much sense.

To emphasize quantitative-interval (vs. quantitative-ratio) comparisons.

If it is the norm (e.g., stock charts).

Perception and cognition

Signal detection



Magnitude estimation

Pre-attentive processing

Selective attention

Change blindness

Gestalt grouping

Questions?

Perception and cognition

Signal detection

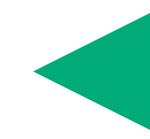
Magnitude estimation

Pre-attentive processing

Selective attention

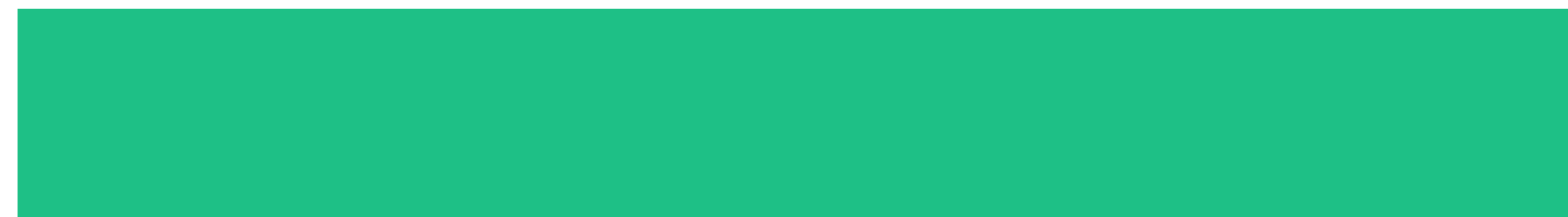
Change blindness

Gestalt grouping

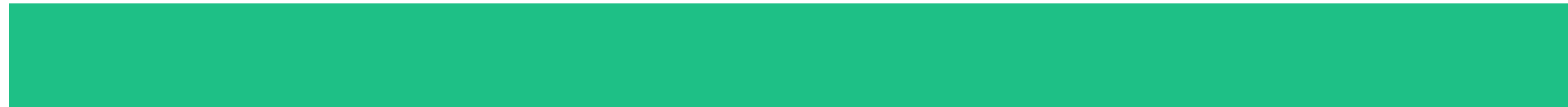


How much longer is the longer bar?

Length encoding



600

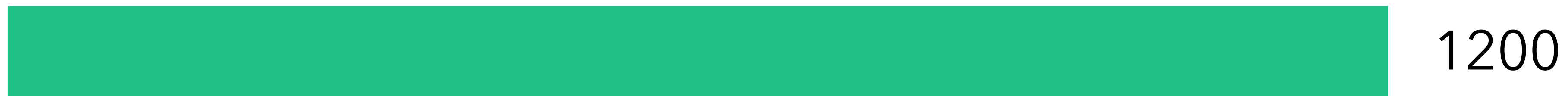


1200

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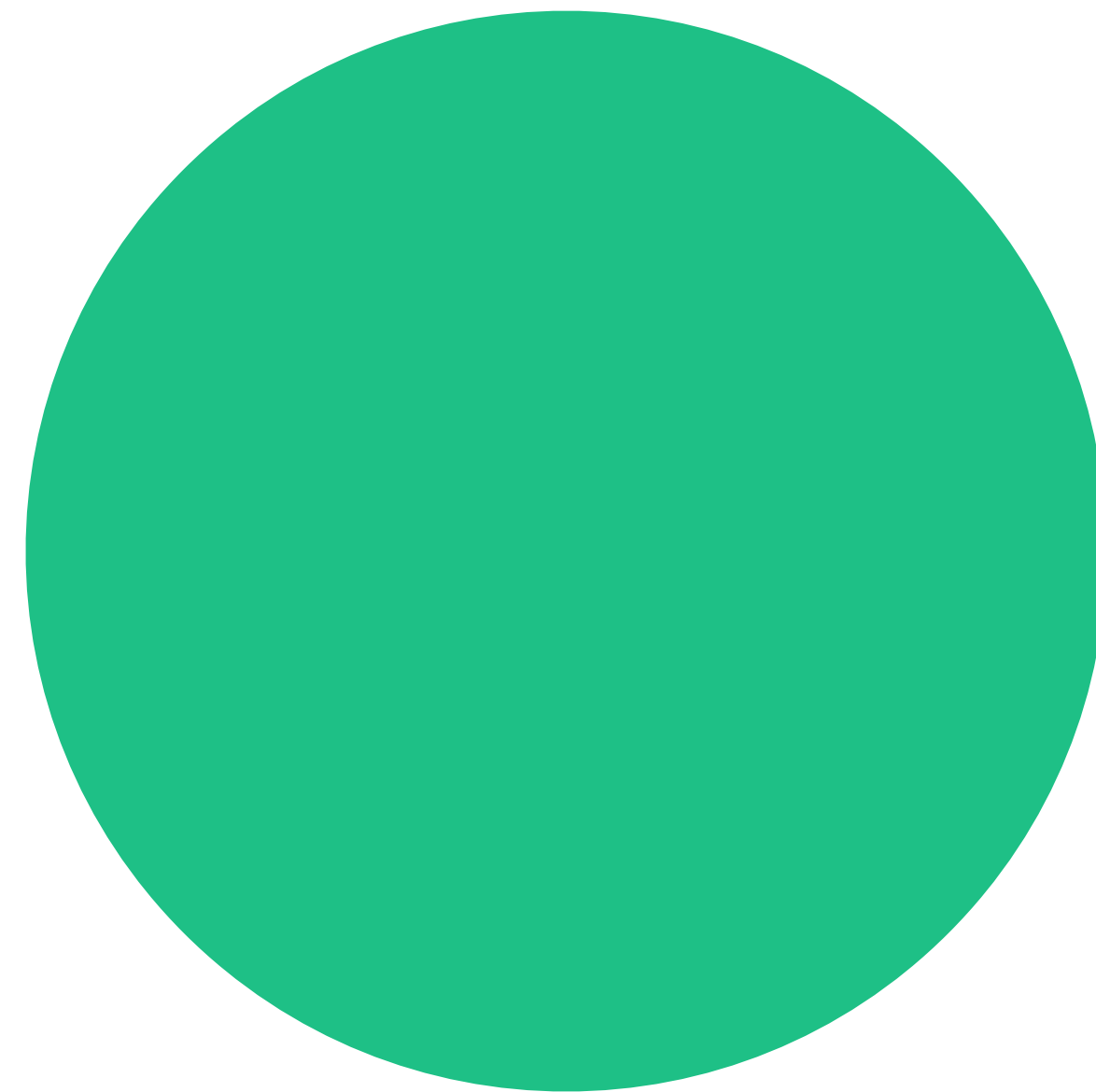
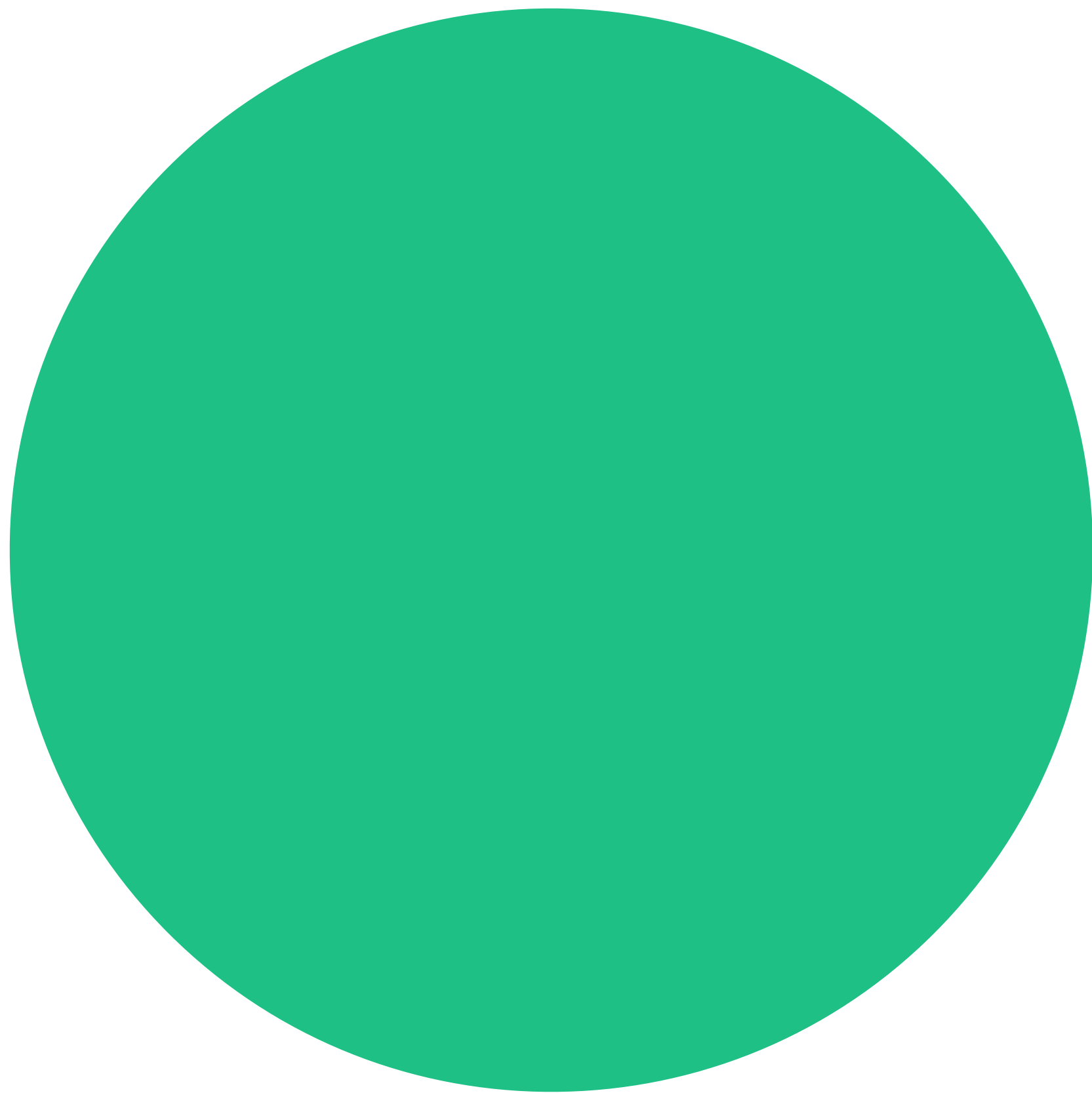
How much longer is the longer bar?

Length encoding



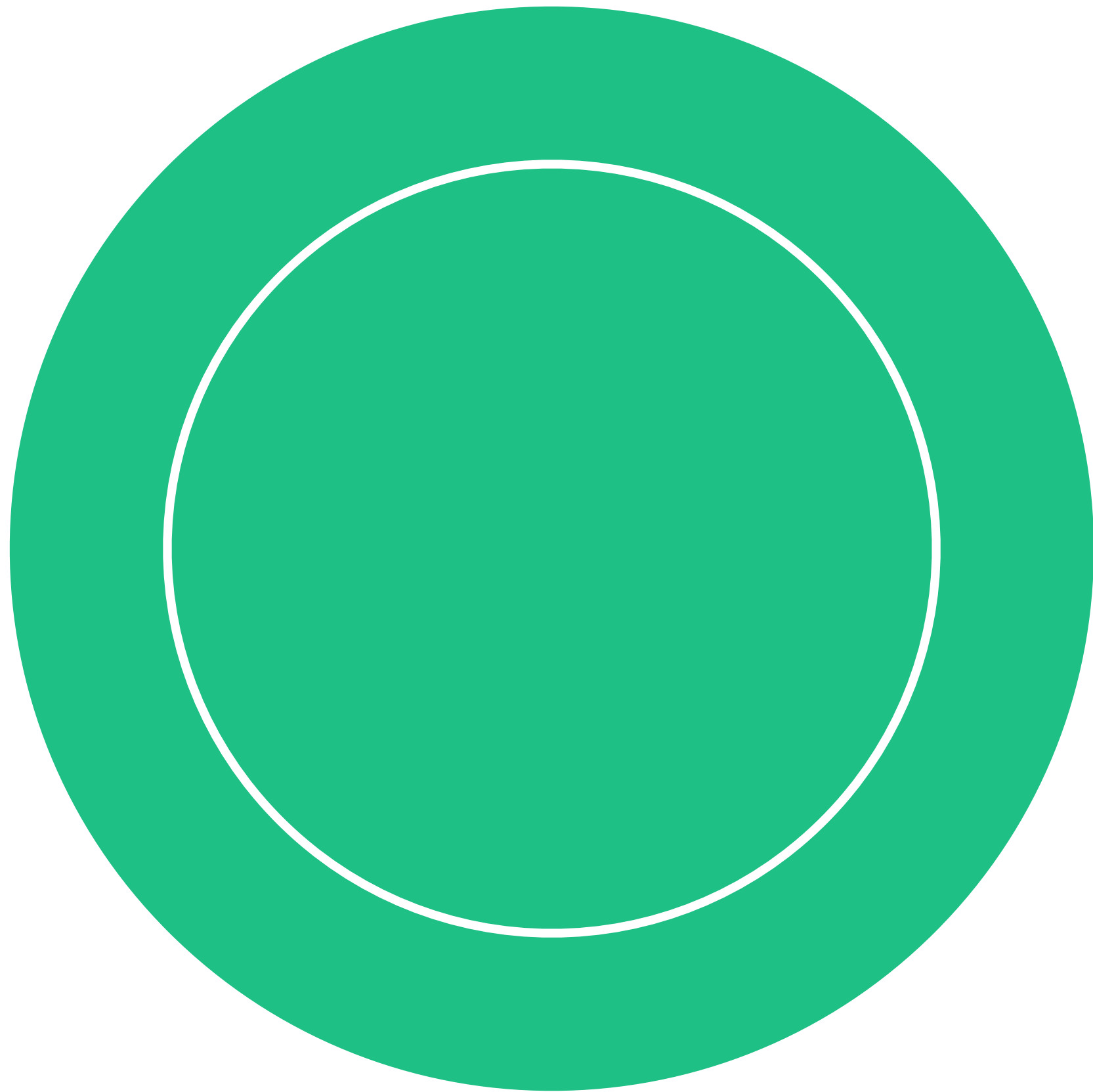
How much larger is the area of the larger circle?

Area encoding



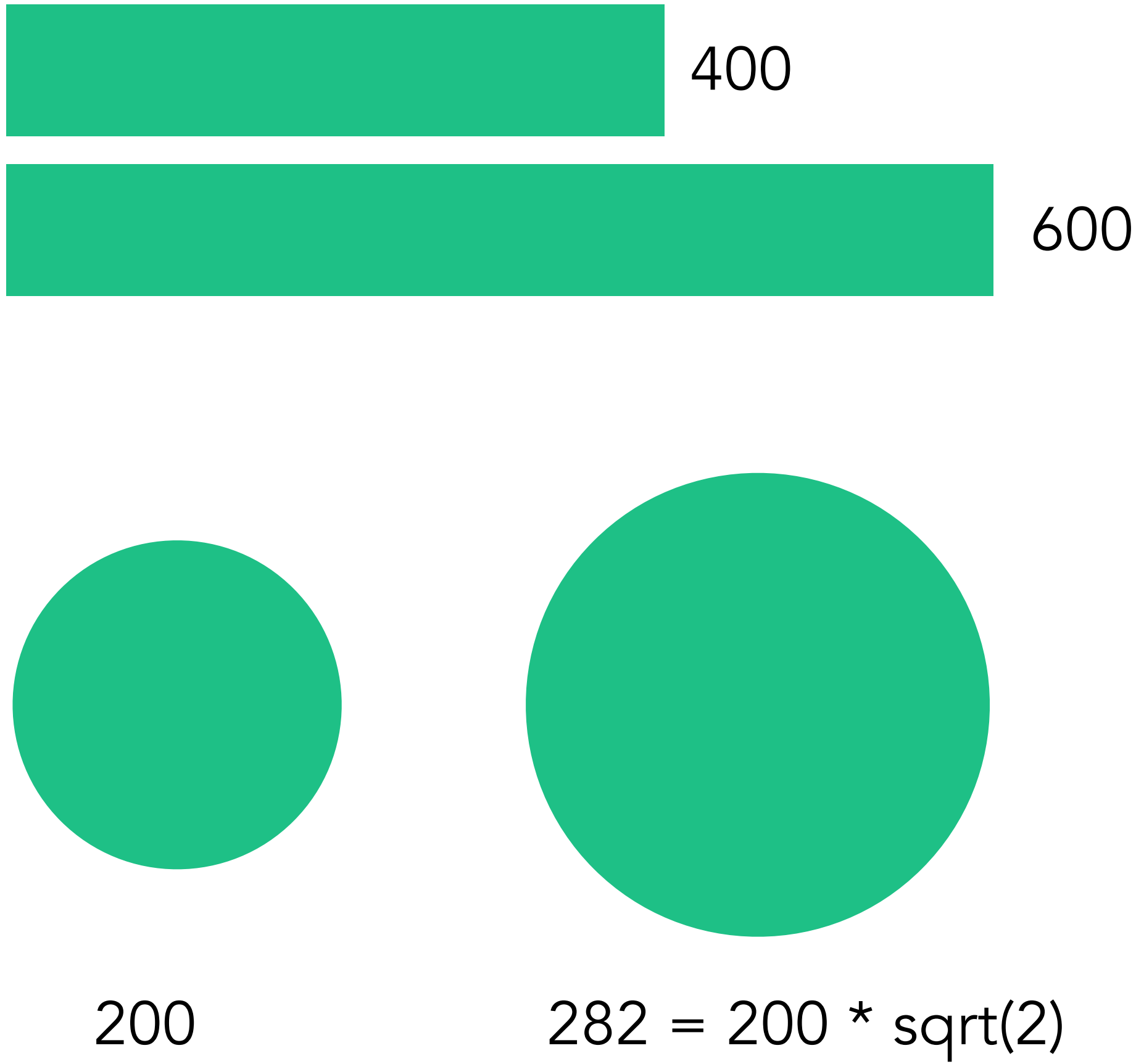
How much larger is the area of the larger circle?

Area encoding



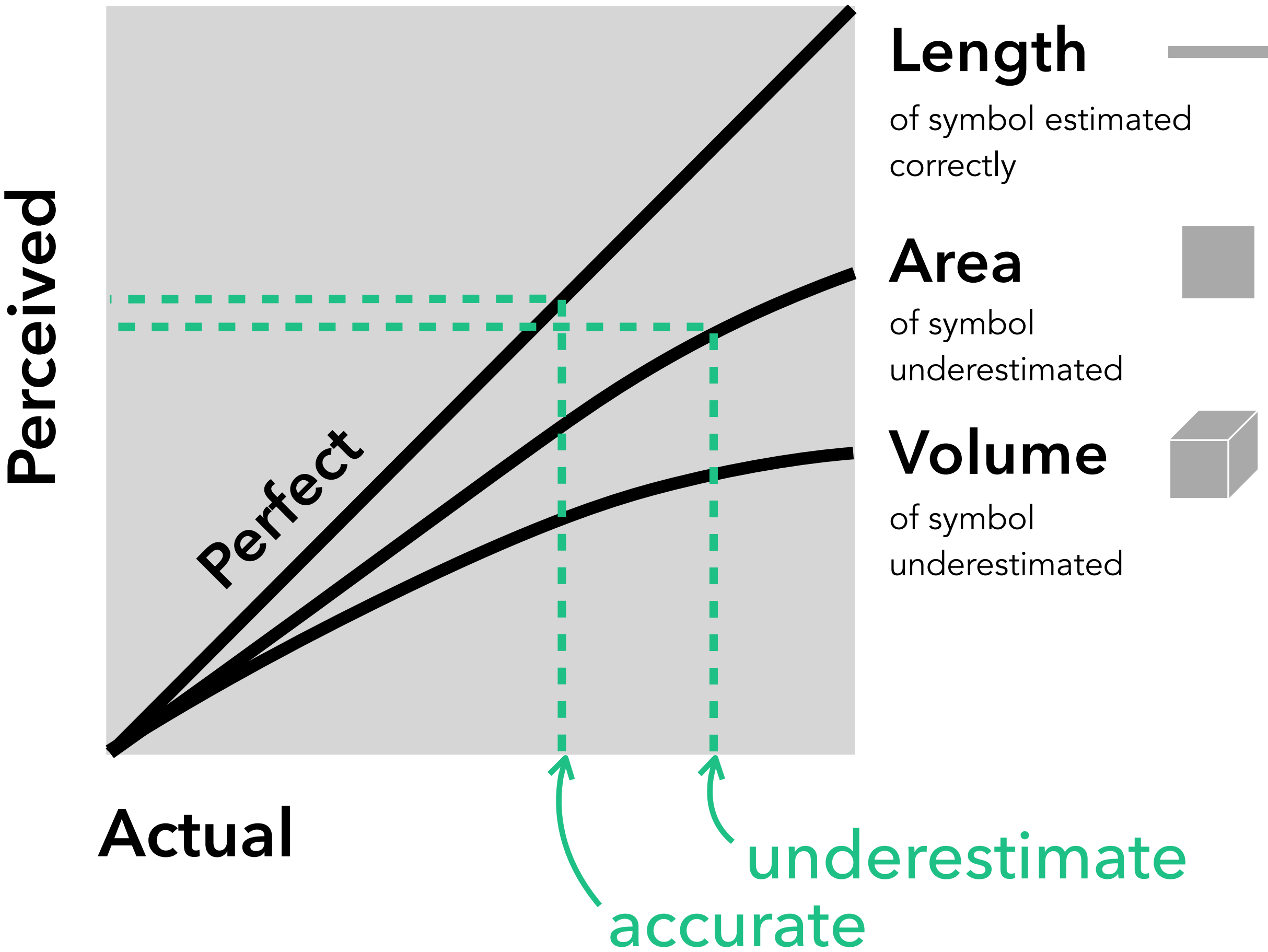
$$\frac{619}{439} = \sim 1.41 = \sqrt{2}$$

Encodings induce bias



Stevens' power law

$$\Phi(I) = kI^a$$



Continuum	a	Stimulus condition
Brightness	0.5	Point source
Smell	0.6	Heptane
Loudness	0.67	Sound pressure of 3000 Hz tone
Visual area	0.7	Projected square
Warmth	0.7	Irradiation of skin, large area
Taste	0.8	Saccharin
Vibration	0.95	Amplitude of 60 Hz on finger
Brightness	1	Point source briefly flashed
Cold	1	Metal contact on arm
Thermal pain	1	Radiant heat on skin
Visual length	1	Projected line
Duration	1.1	White-noise stimuli
Vocal effort	1.1	Vocal sound pressure
Lightness	1.2	Reflectance of gray papers
Taste	1.3	Sucrose
Warmth	1.3	Irradiation of skin, small area
Electric shock	3.5	Current through fingers

Stevens' power law

$$\Phi(I) = kI^a$$

Exponent

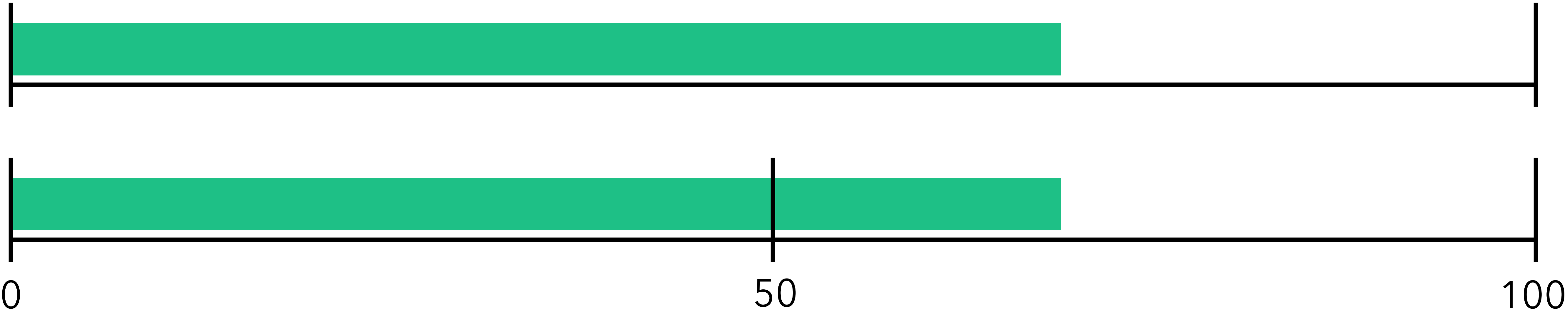
Perceived

Actual

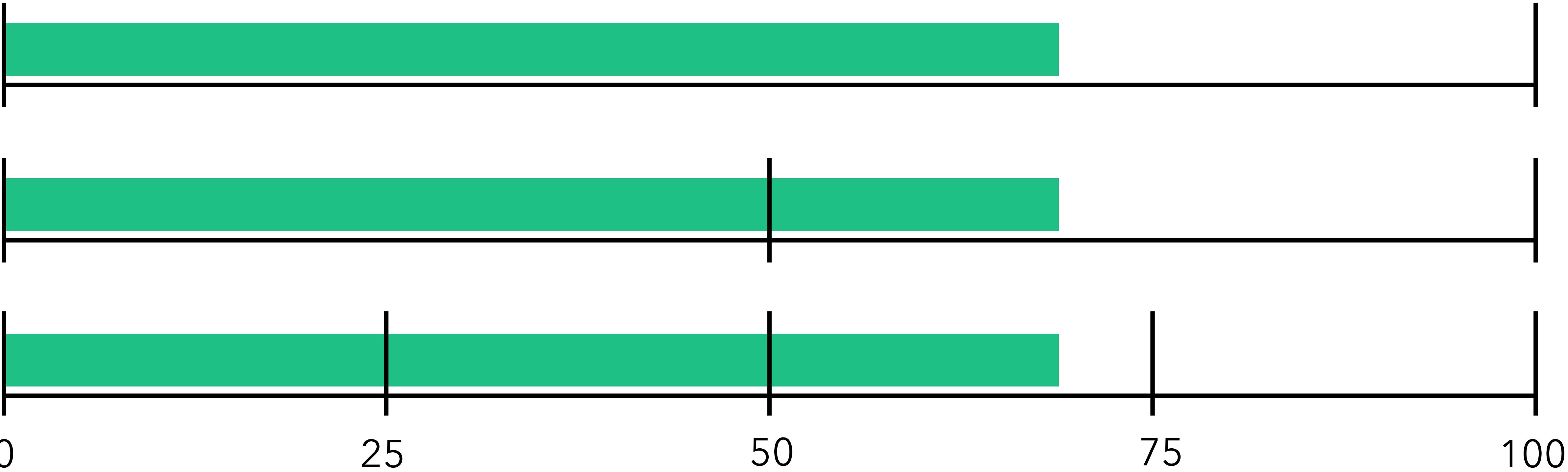
Reference lines can help



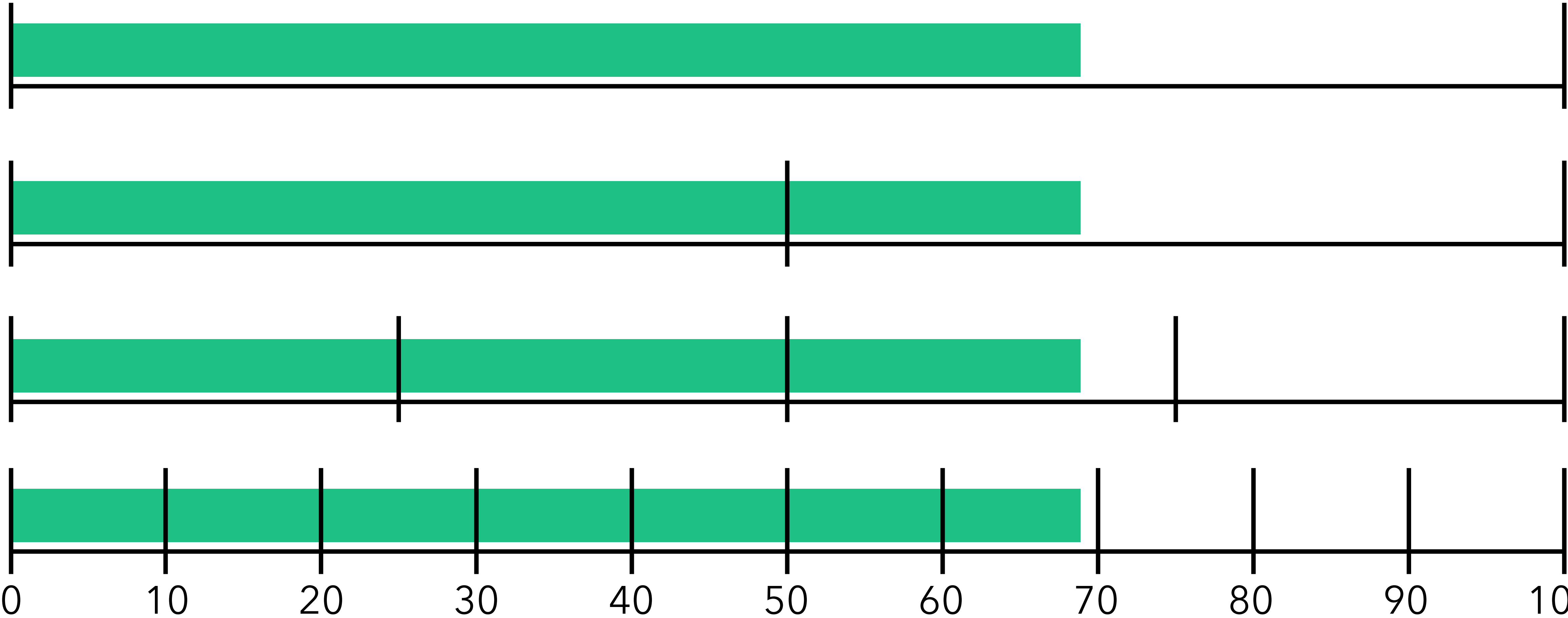
Reference lines can help



Reference lines can help



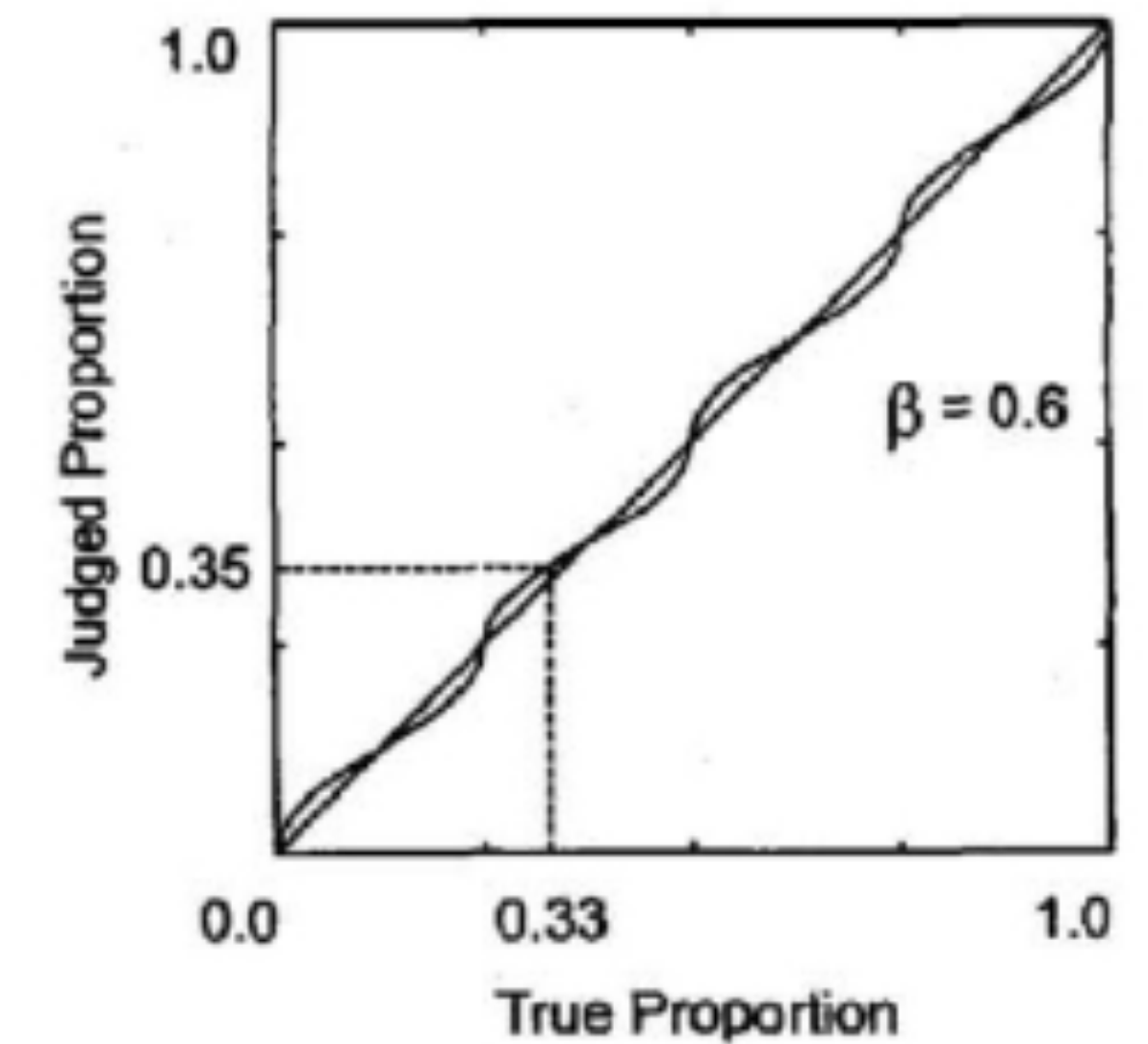
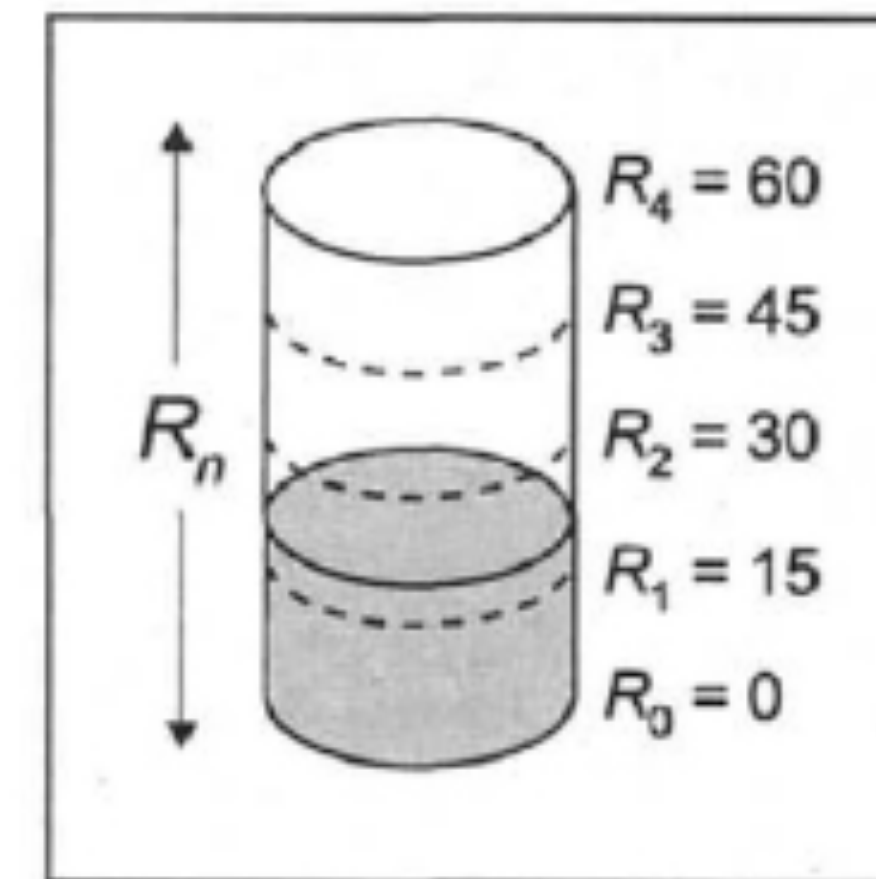
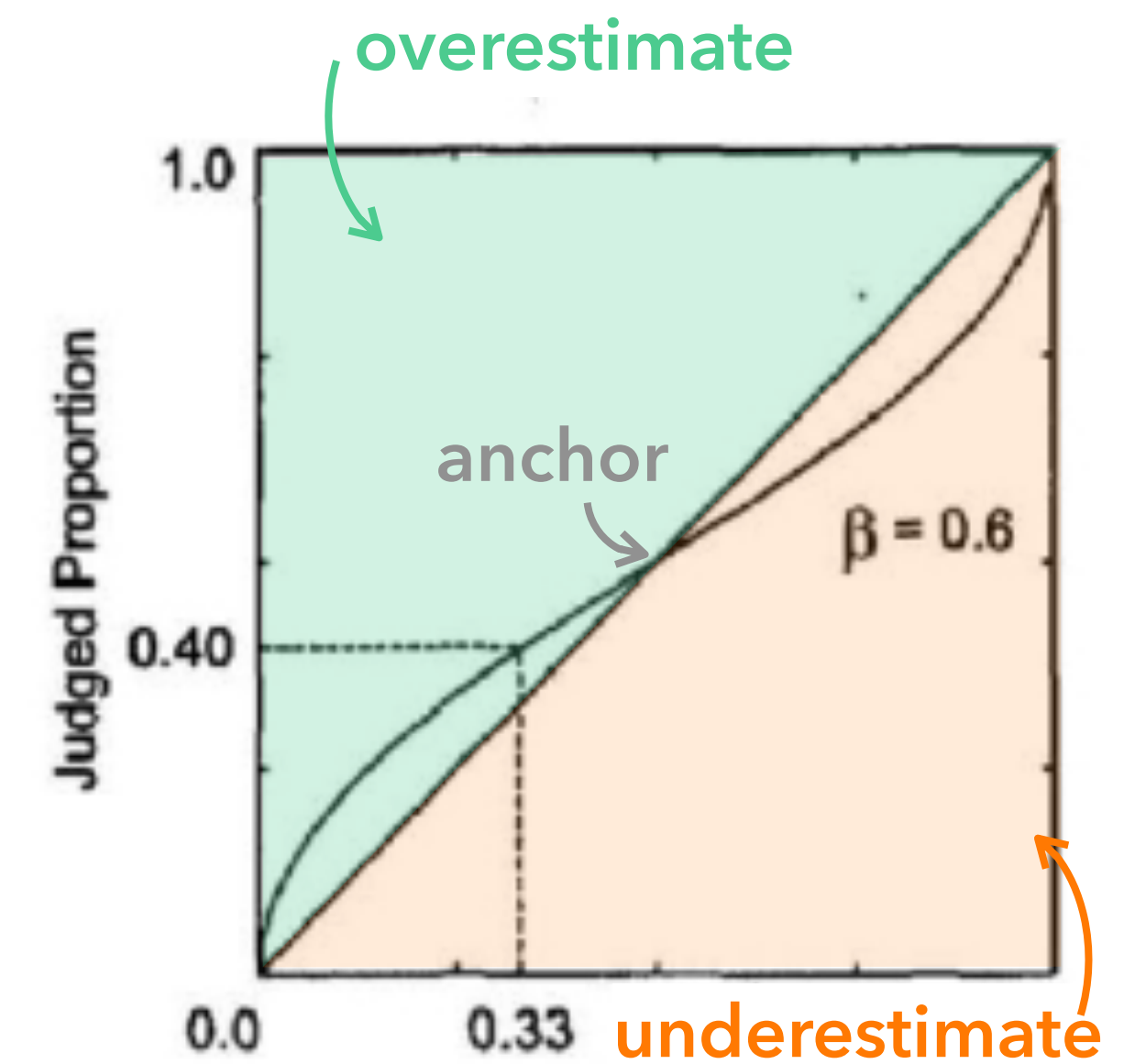
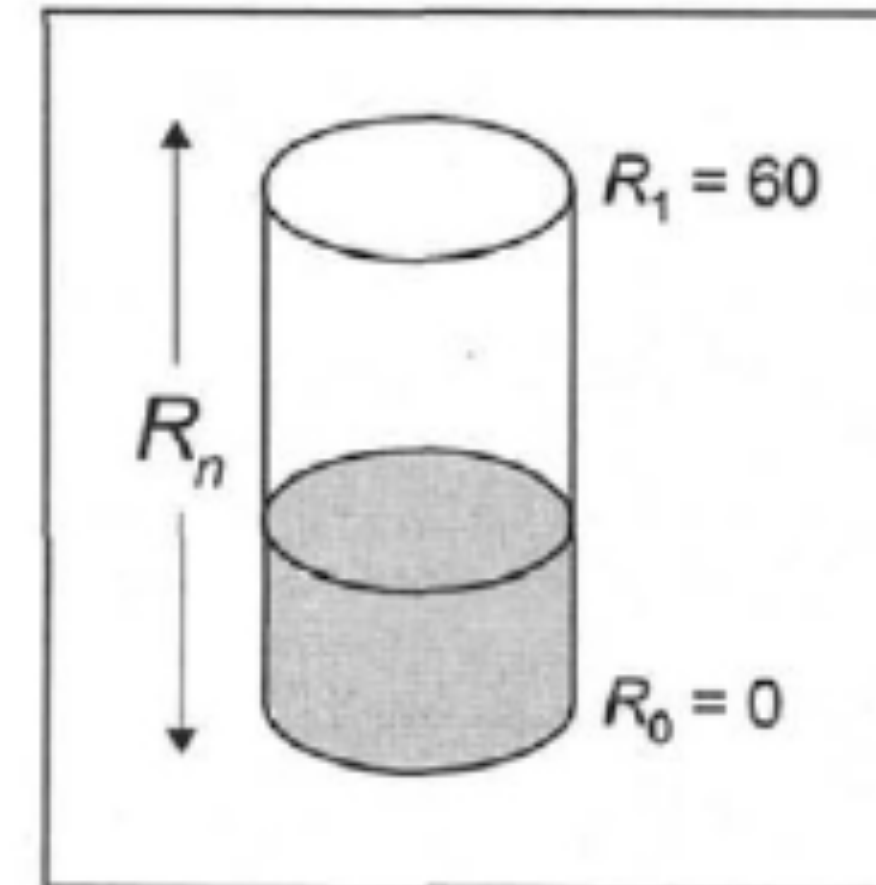
Reference lines can help....?



Reference lines can help....?

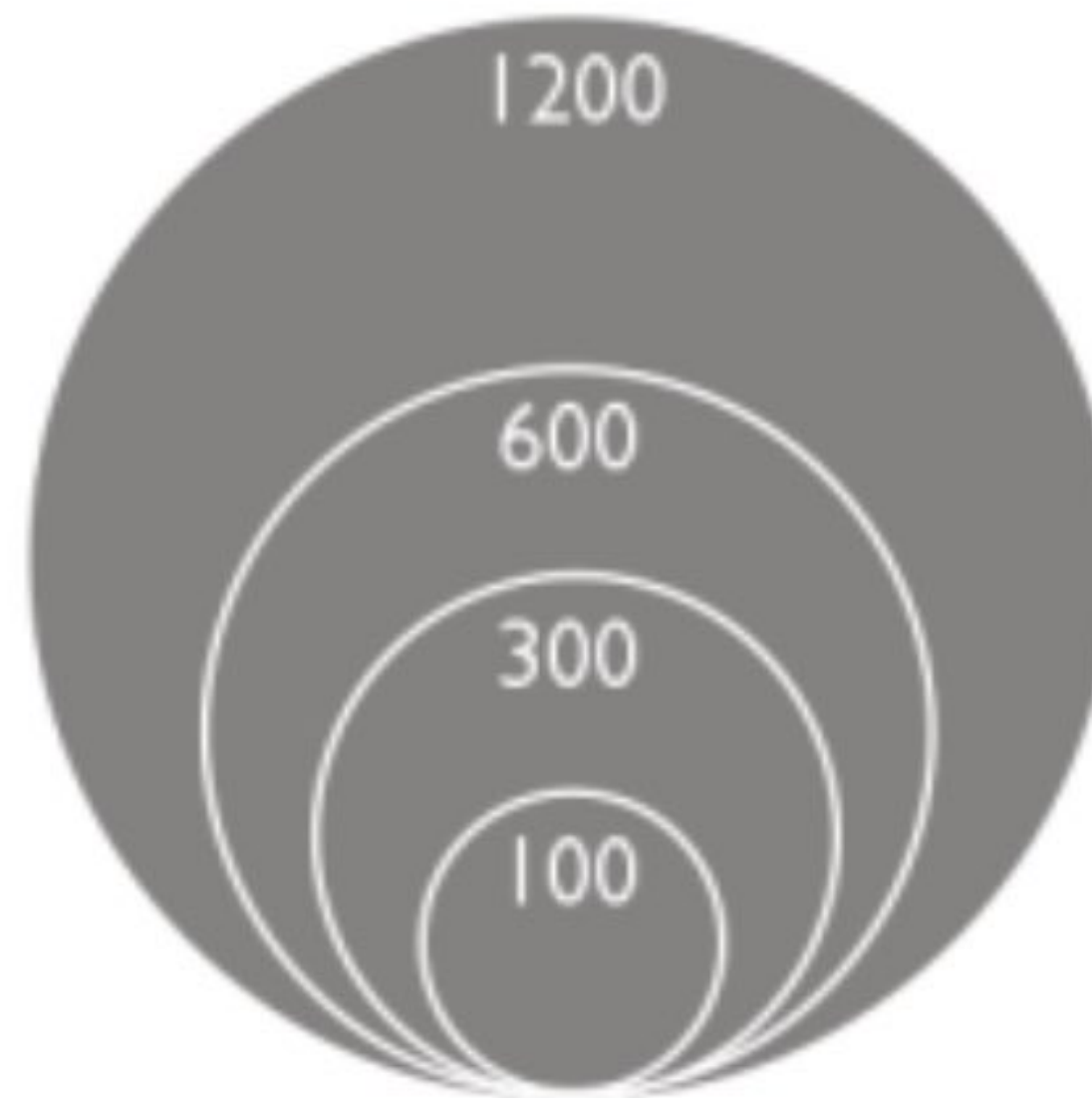
Also induce bias

...but generally **decrease** error

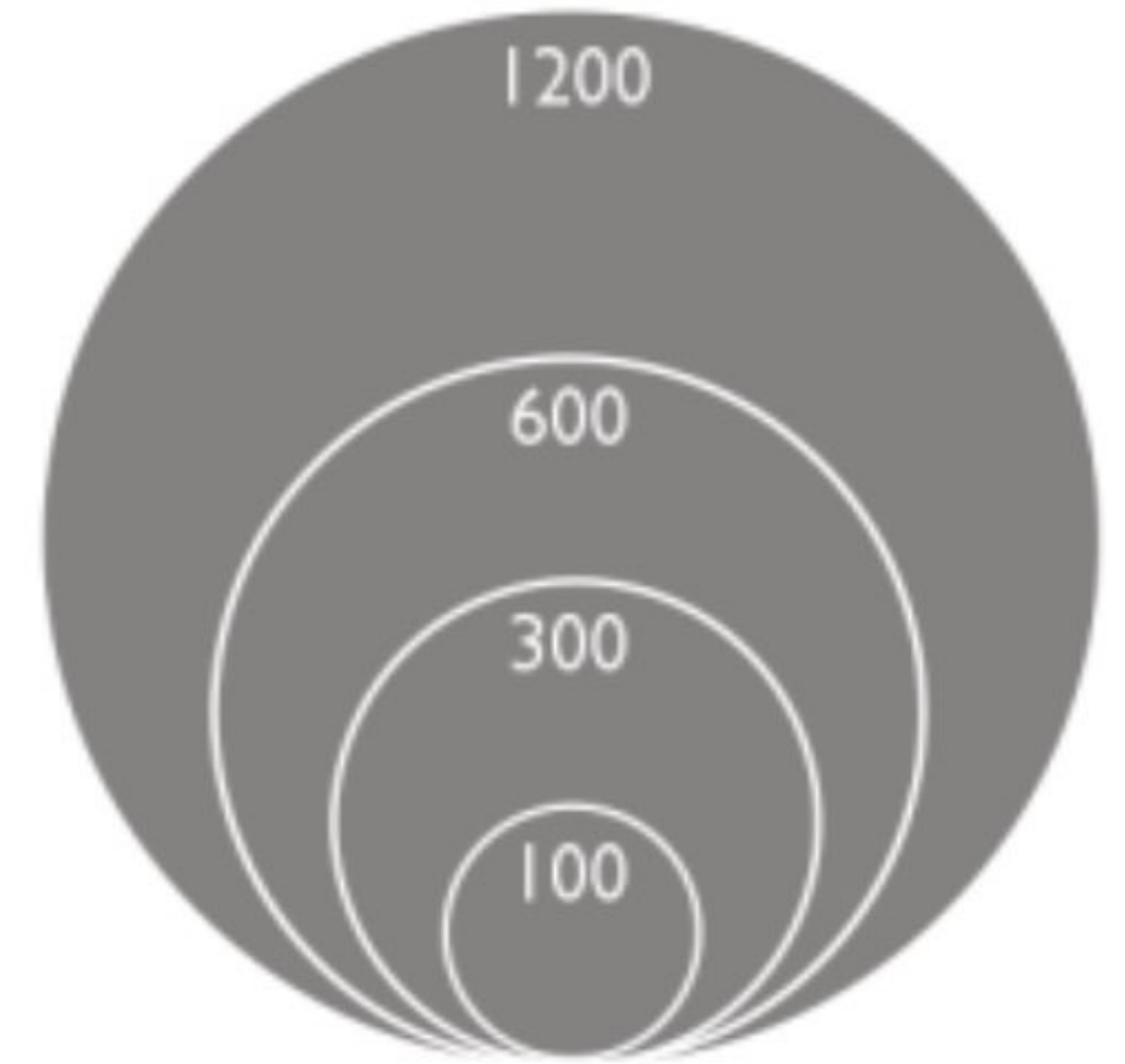


Apparent scaling

If you want to convey that A is 2x B, you should actually size area A to be **more** than 2x area B



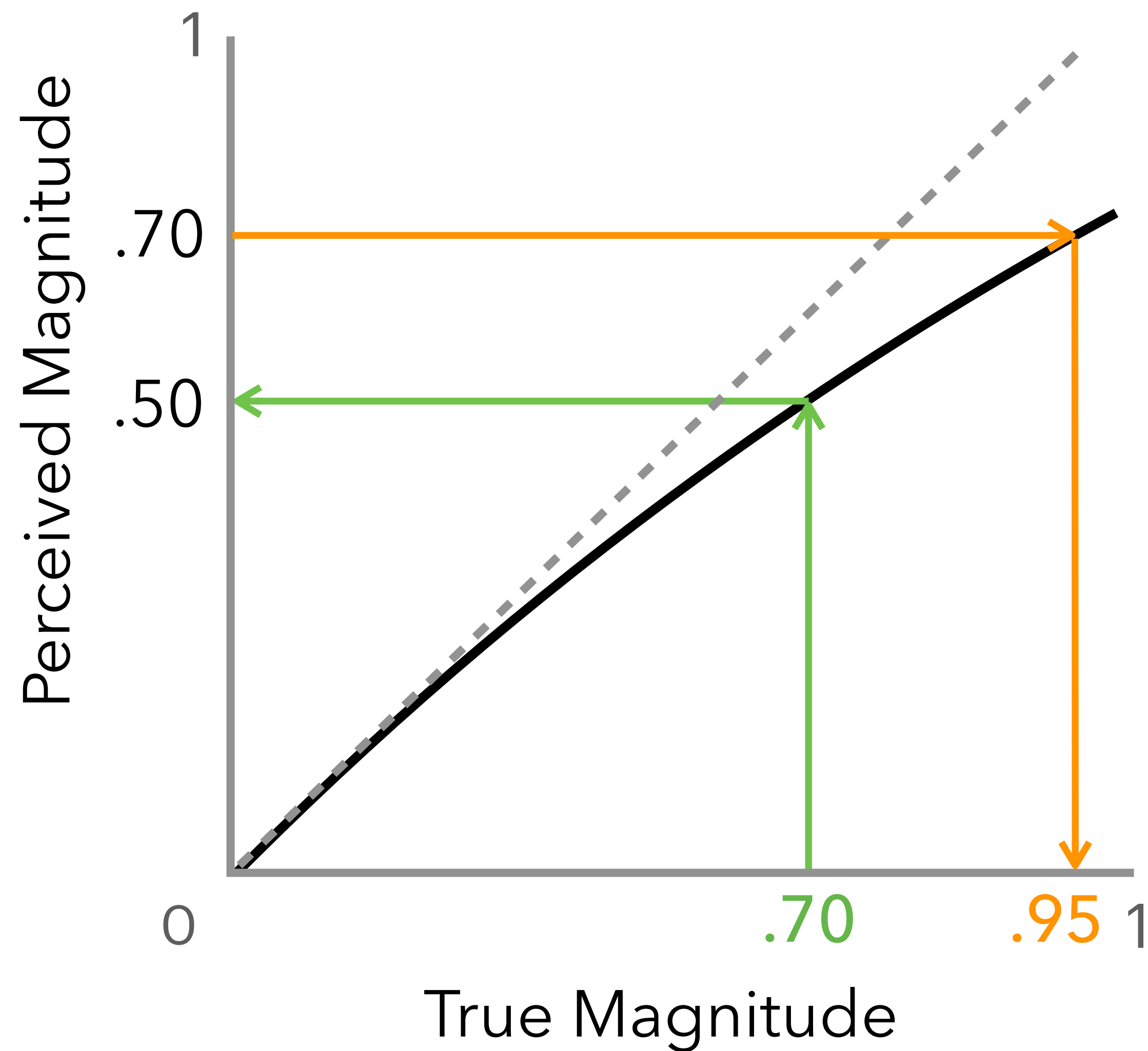
Absolute Scaling



Apparent Scaling

(Flannery's Compensation)

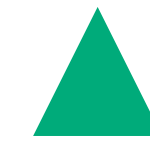
Invert the perceptual function



Perception and cognition

Signal detection

Magnitude estimation



Pre-attentive processing

Selective attention

Change blindness

Gestalt grouping

Questions?

Perception and cognition

Signal detection

Magnitude estimation

Pre-attentive processing

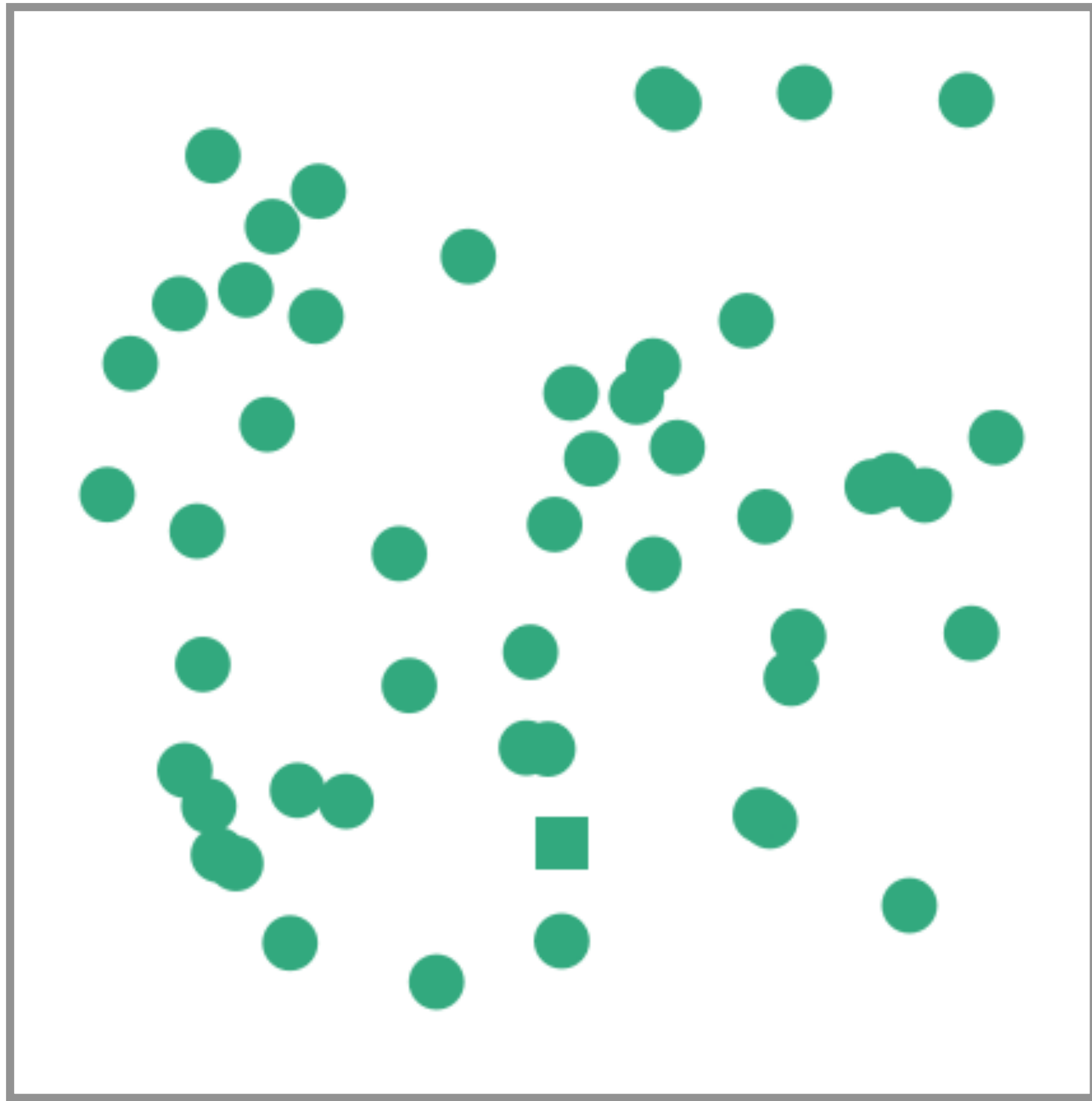
Selective attention

Change blindness

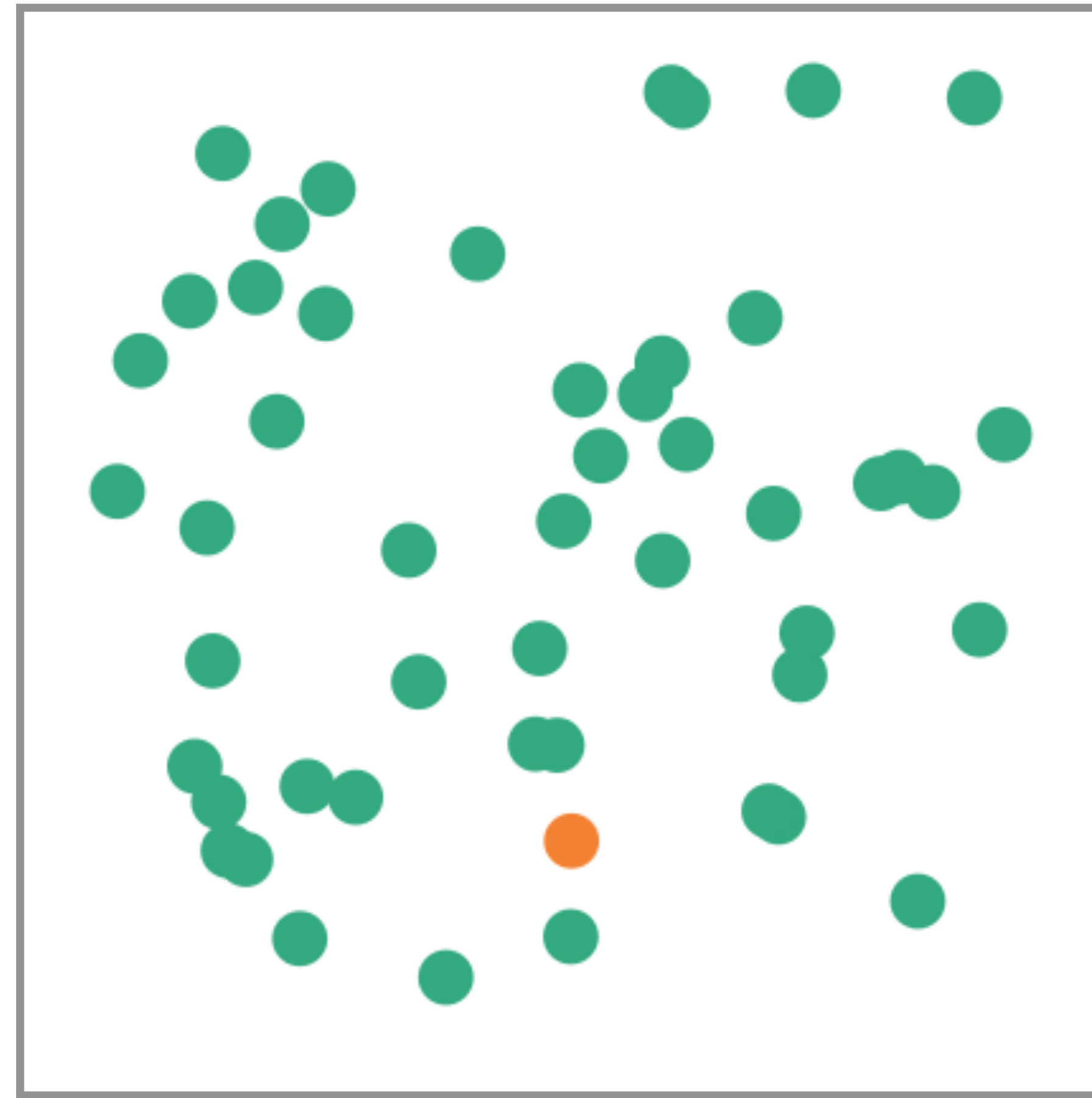
Gestalt grouping



Popout and salience



Visual pop-out: **shape**



Visual pop-out: **color**

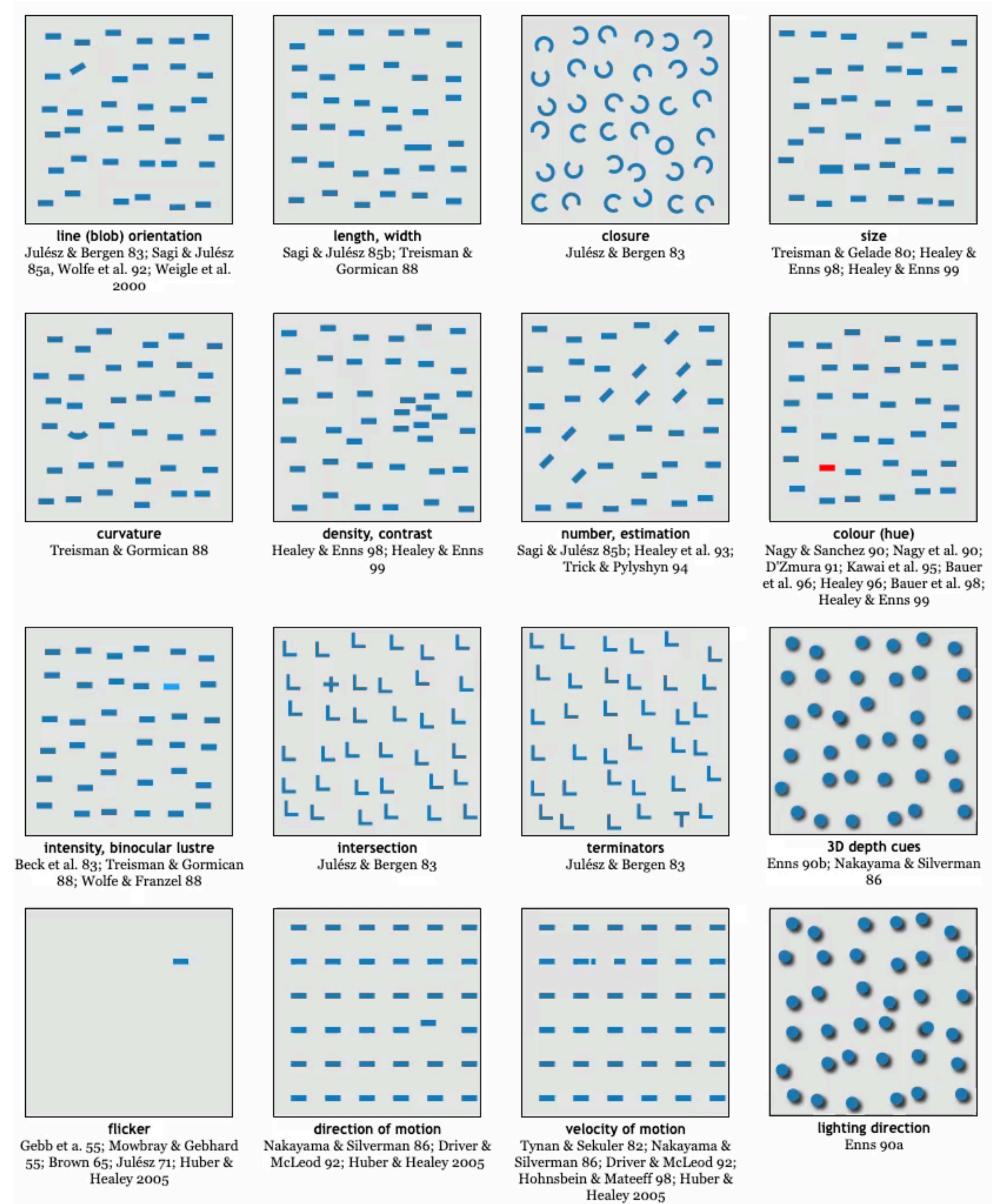
Pre-attentive processing

Pre-attentive

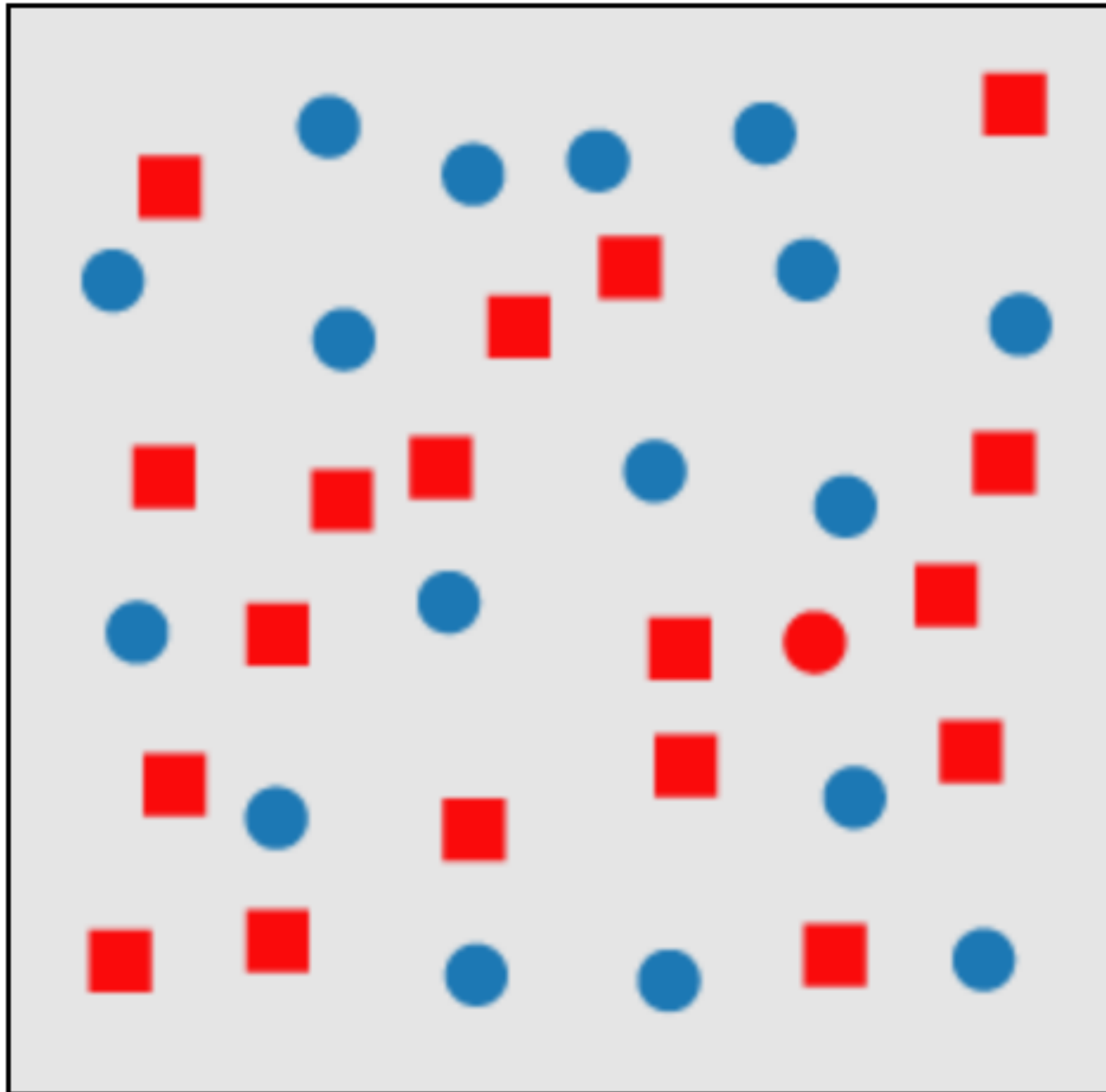
immediately recognize variation
with little or no conscious effort
($<200\text{--}250\text{ ms}$)

Attentive

Takes some delicate effort to
perceive differences.



Feature conjunctions: Is a **red circle** present?



Conjunctions are not pre-attentive
except for spatial conjunctions:

Motion & 3D disparity

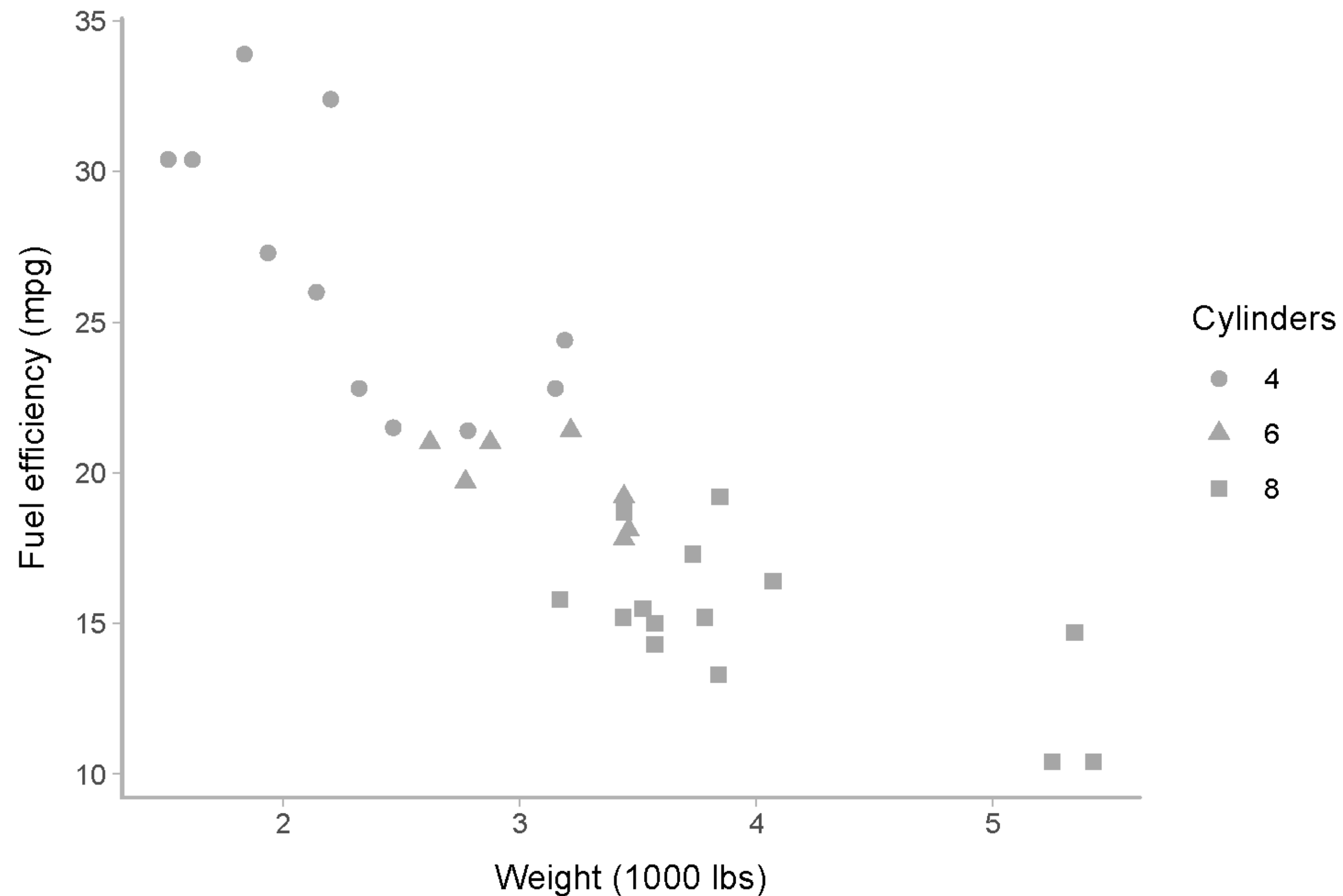
Motion & color

Motion & shape

3D disparity & color

3D disparity & shape

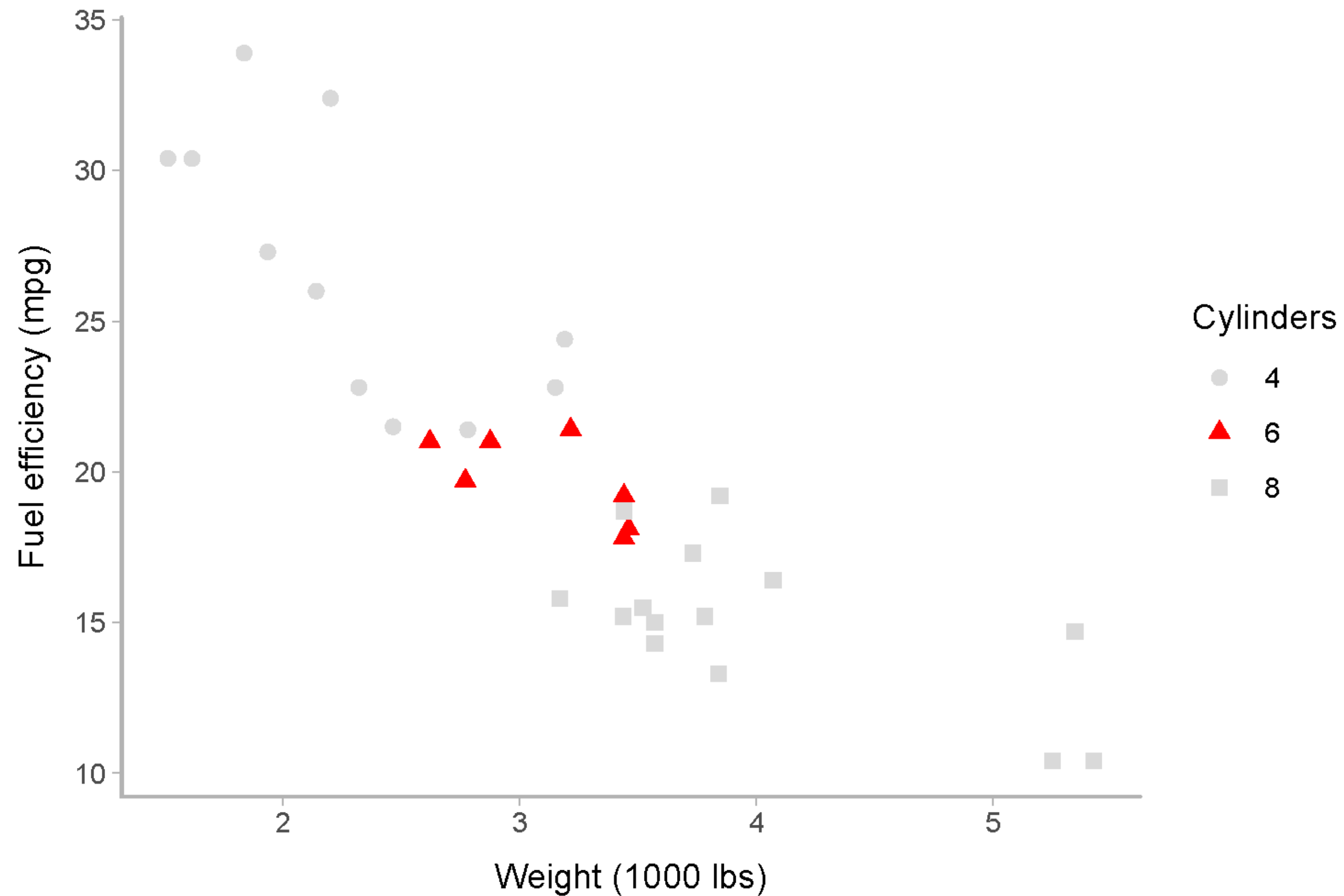
Popout and salience



What do people see **first**?

What can people see **separately**? (layering)

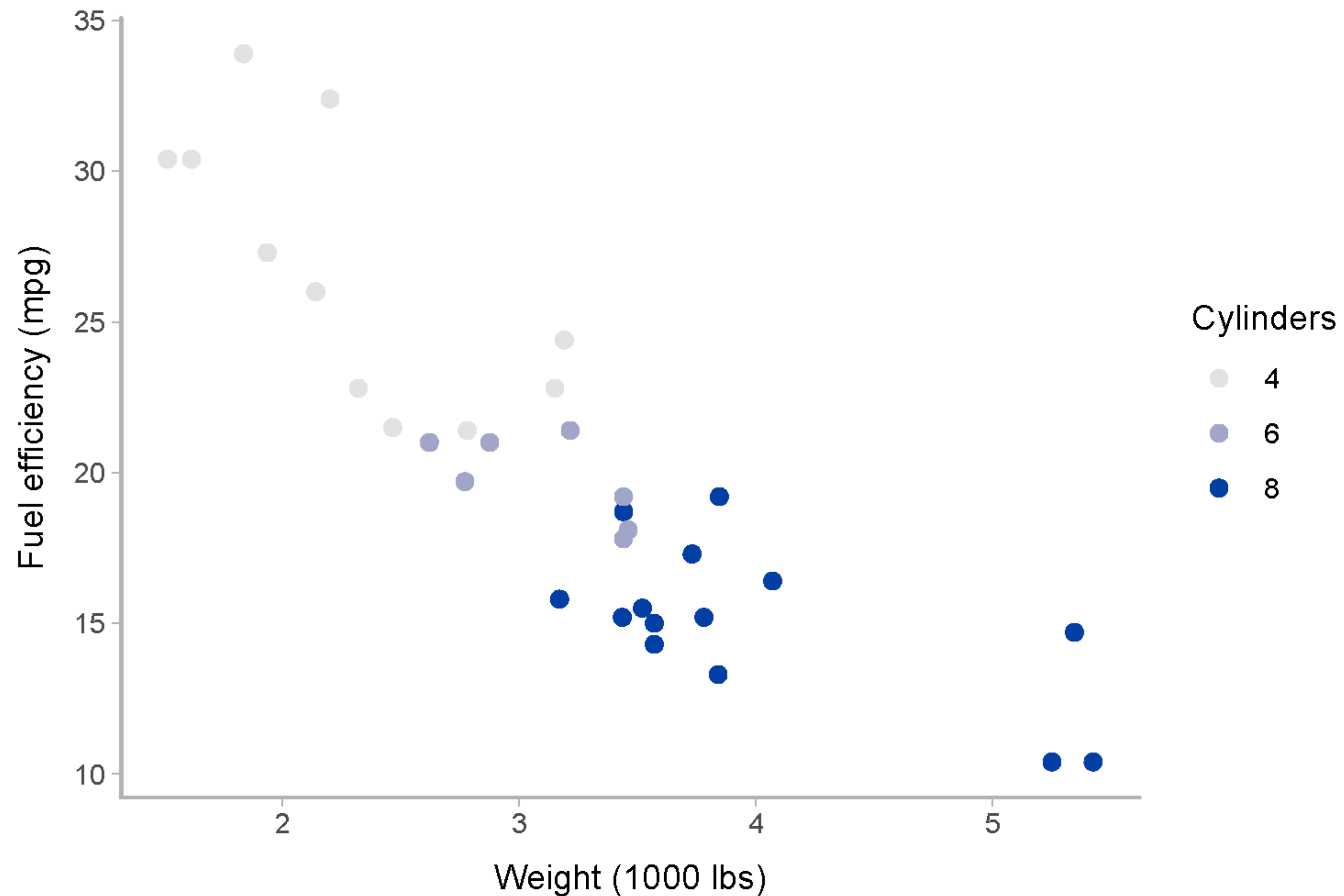
Popout and salience



What do people see **first**?

What can people see **separately**? (layering)

Popout and salience



What do people see **first**?

What can people see **separately**? (layering)

Question?

Perception and cognition

Signal detection

Magnitude estimation

Pre-attentive processing

Selective attention

Change blindness

Gestalt grouping



Questions?

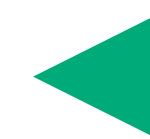
Perception and cognition

Signal detection

Magnitude estimation

Pre-attentive processing

Selective attention



Change blindness

Gestalt grouping

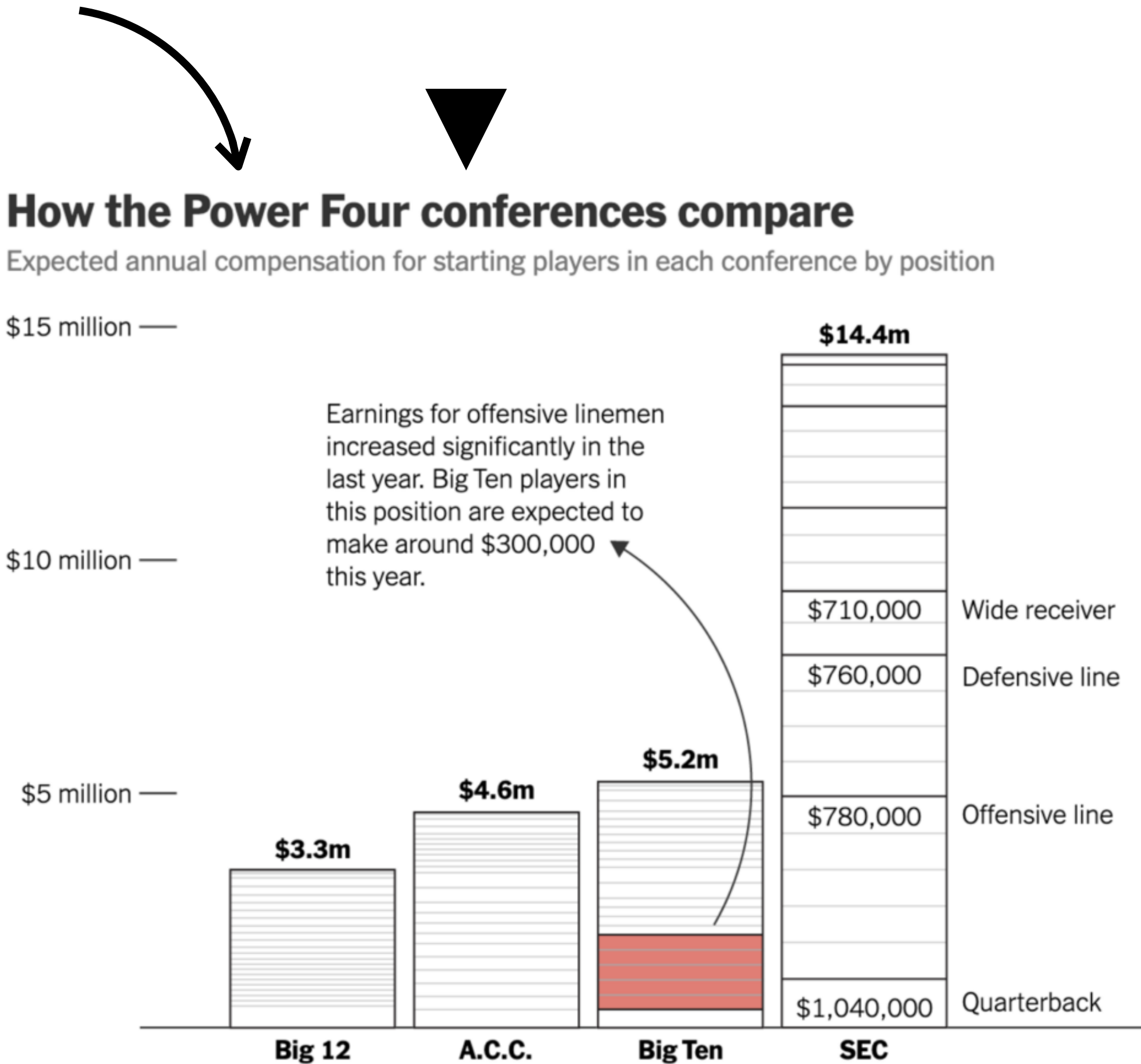


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Lecture & presentation

I have been using annotations and gestures for figures to guide your attention all the time...

Voice over for a visualization
Annotations for a visualization



Source: [Opendorse](#). Data is based on N.I.L. transactions disclosed through or processed by Opendorse between July 1, 2021, and June 30, 2024. • Note: To be included in the calculations players' earnings must rank in the top 10 at their position.

Application of selective attention...

Guiding attention: Design visual elements (color, size, position) to direct attention to key information.

Reducing information overload: Highlight relevant data while minimizing distractions from less important details.

Supporting task goals: Align visual emphasis with user tasks.

Interactive exploration: Enable users to filter or zoom in on data.

Perception and cognition

Signal detection

Magnitude estimation

Pre-attentive processing

Selective attention

Change blindness

Gestalt grouping



Questions?

Perception and cognition

Signal detection

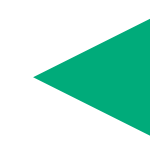
Magnitude estimation

Pre-attentive processing

Selective attention

Change blindness

Gestalt grouping



Did You **See** It?



Application of change blindness...

Understanding user limitations: Users may overlook subtle changes.

Designing effective cues: Use color shifts, motion, or animations.

Enhancing situational awareness: Emphasize key changes to prevent important data from being missed.

Real-time systems: Dashboards where rapid data updates occur

Reducing cognitive load: Clear visual contrasts help users focus.

Perception and cognition

Signal detection

Magnitude estimation

Pre-attentive processing

Selective attention

Change blindness

Gestalt grouping



Questions?

Perception and cognition

Signal detection

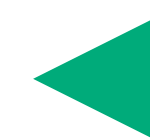
Magnitude estimation

Pre-attentive processing

Selective attention

Change blindness

Gestalt grouping



Gestalt grouping

A theory that explains how humans perceive objects as certain **groups**

I feel it's more useful for user **interface design** than visualization design...

e.g. radio button group

But could explain some vis **perception**

Size (optional)

☒ Small

☐ Medium

☐ Large

Size (required)

☒ Small

☐ Medium

☐ Large

Select a product size.

Size *

☒ Small

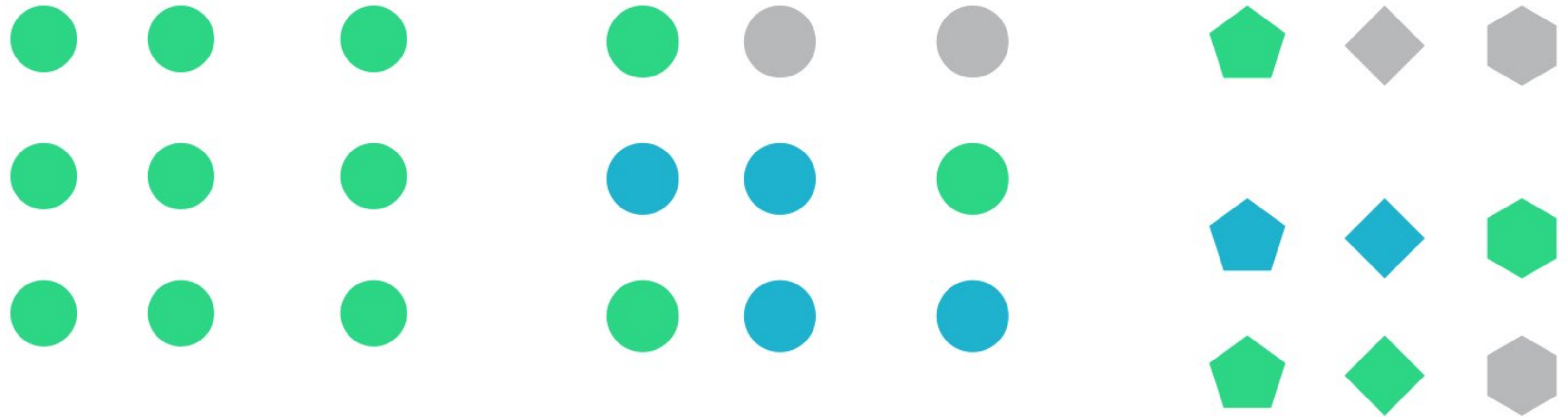
☐ Medium

☐ Large

Select a product size.

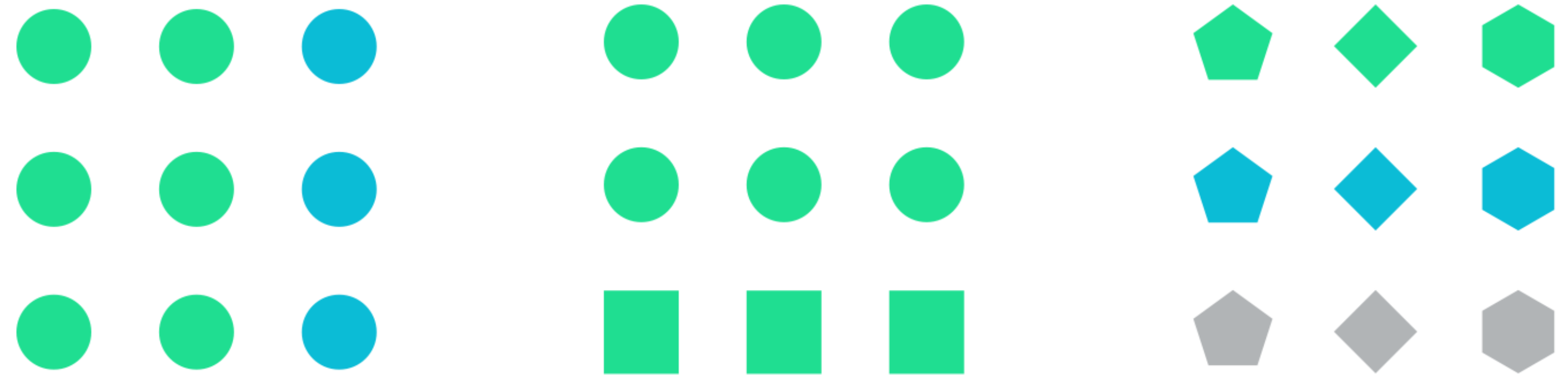
Proximity

Spatial or temporal **grouping** of elements may induce us to perceive a collective or totality.



Similarity

Eyes group similar elements into collective entities or totalities, depending upon relationships of form, color, size or brightness.



Symmetry

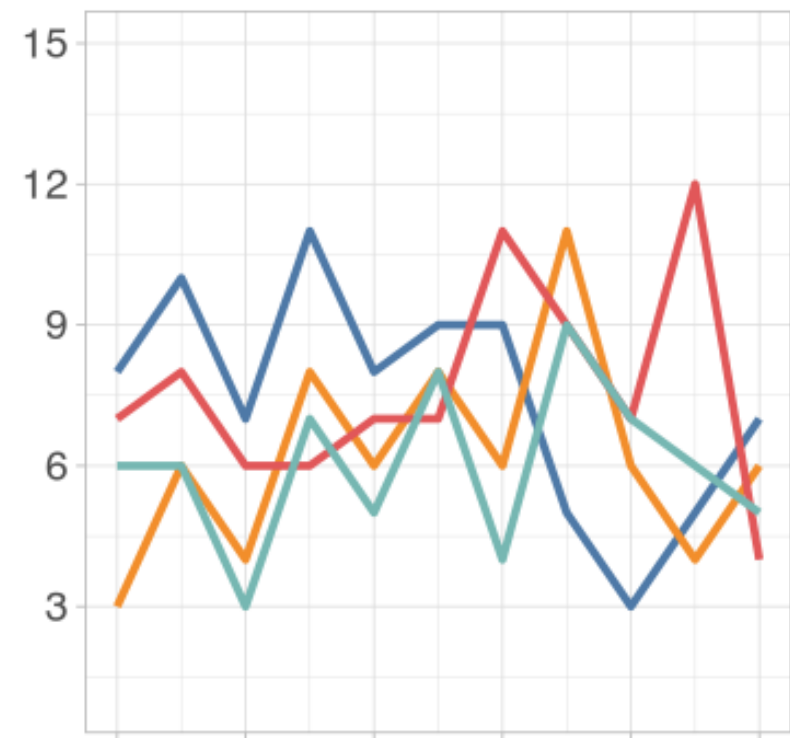
Symmetrical elements are perceived collectively, even in spite of distance.



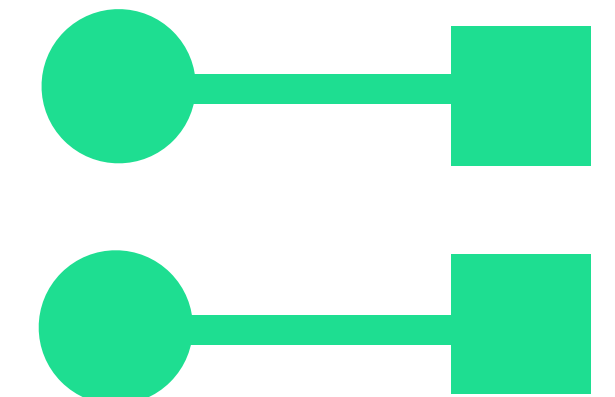
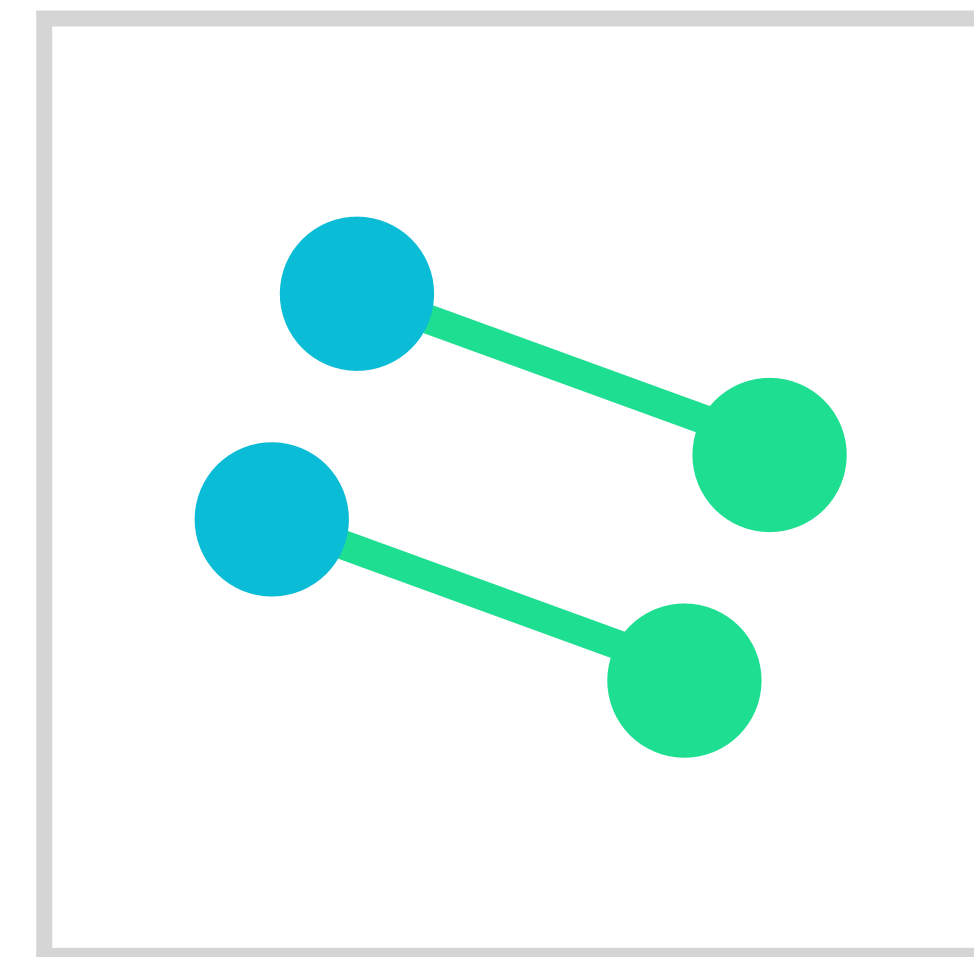
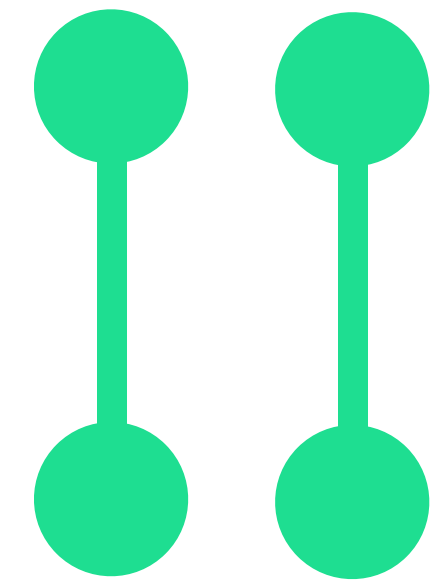
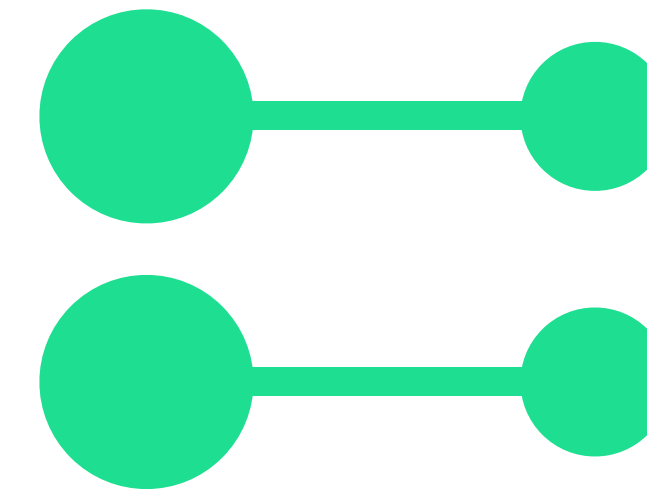
Connectedness

Eyes perceive elements that are visually **connected** as being more related than elements that are not connected

overrules proximity, size, color, shape, ...

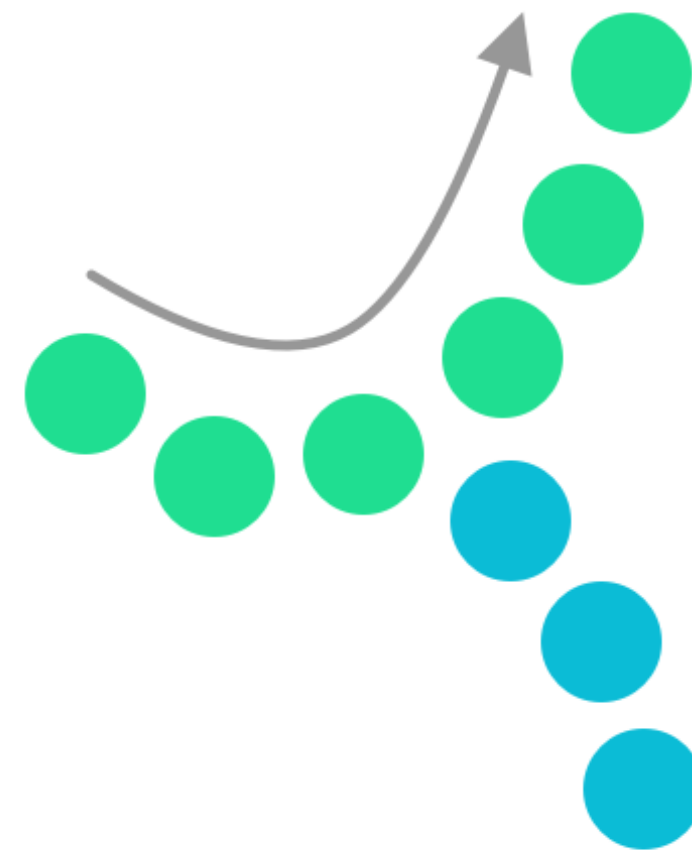


line chart



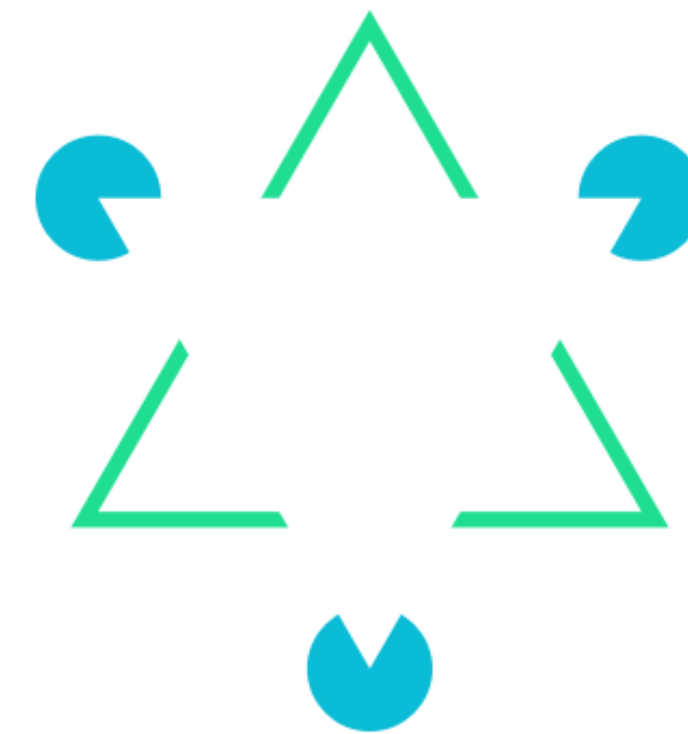
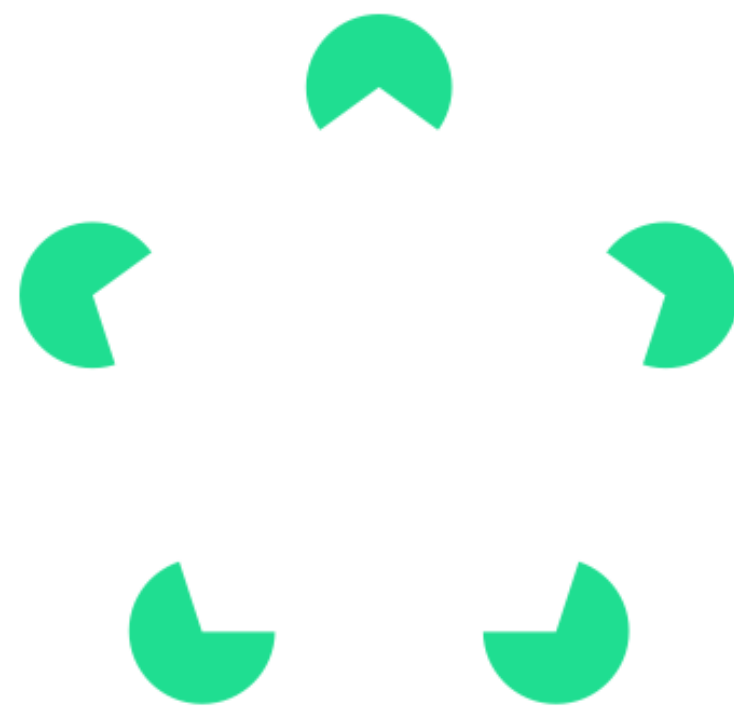
Continuity

Our eyes are more likely to see **continuous and smooth flowing lines** rather than broken or jagged ones



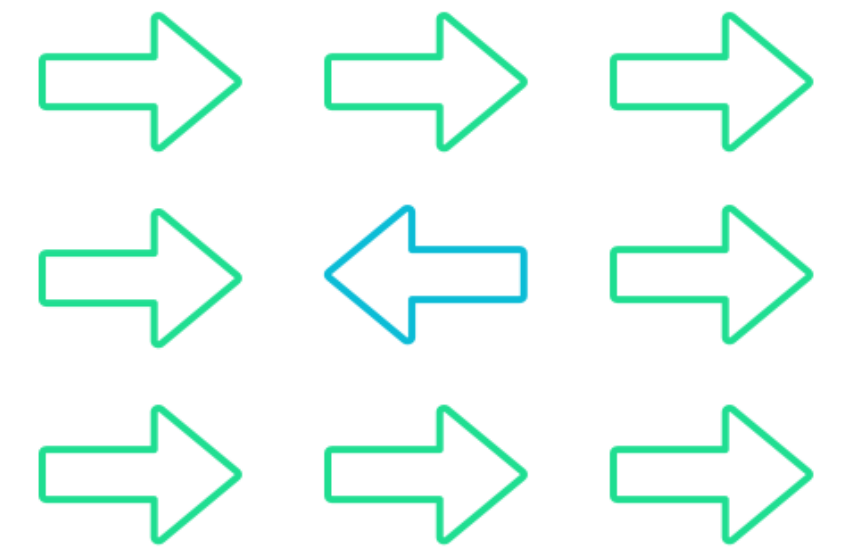
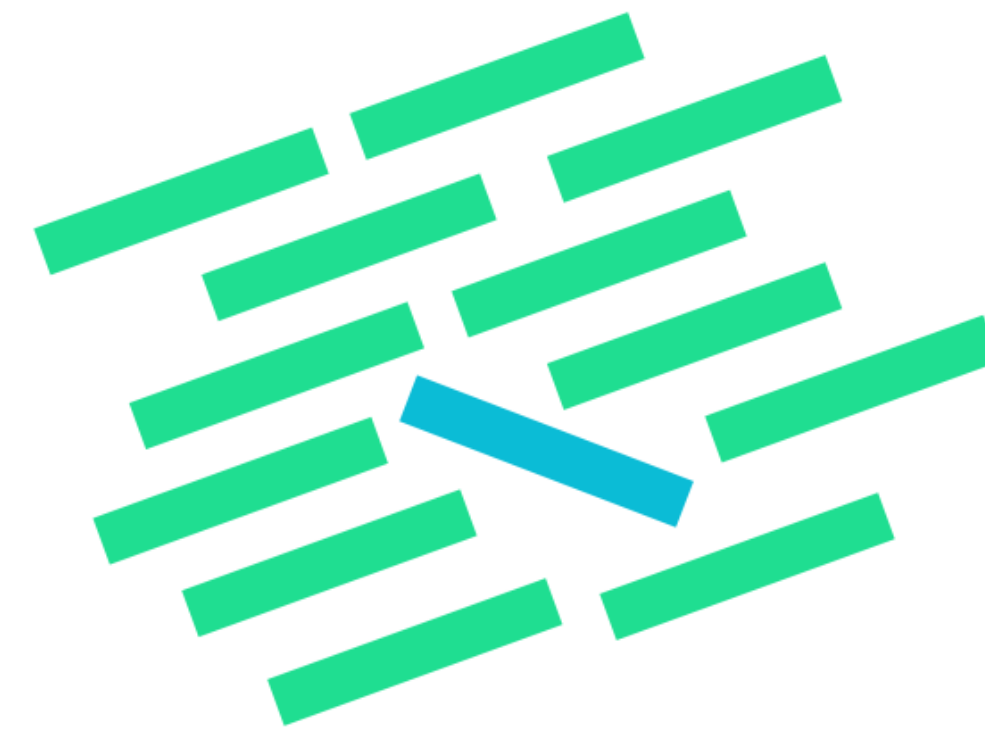
Closure

We may experience elements it does not perceive through sensation, in order to complete a regular figure (that is, to increase regularity).



Common fate

Elements with the same **moving** direction are perceived as a collective or unit.





Perception and cognition

Signal detection

Magnitude estimation

Pre-attentive processing

Selective attention

Change blindness

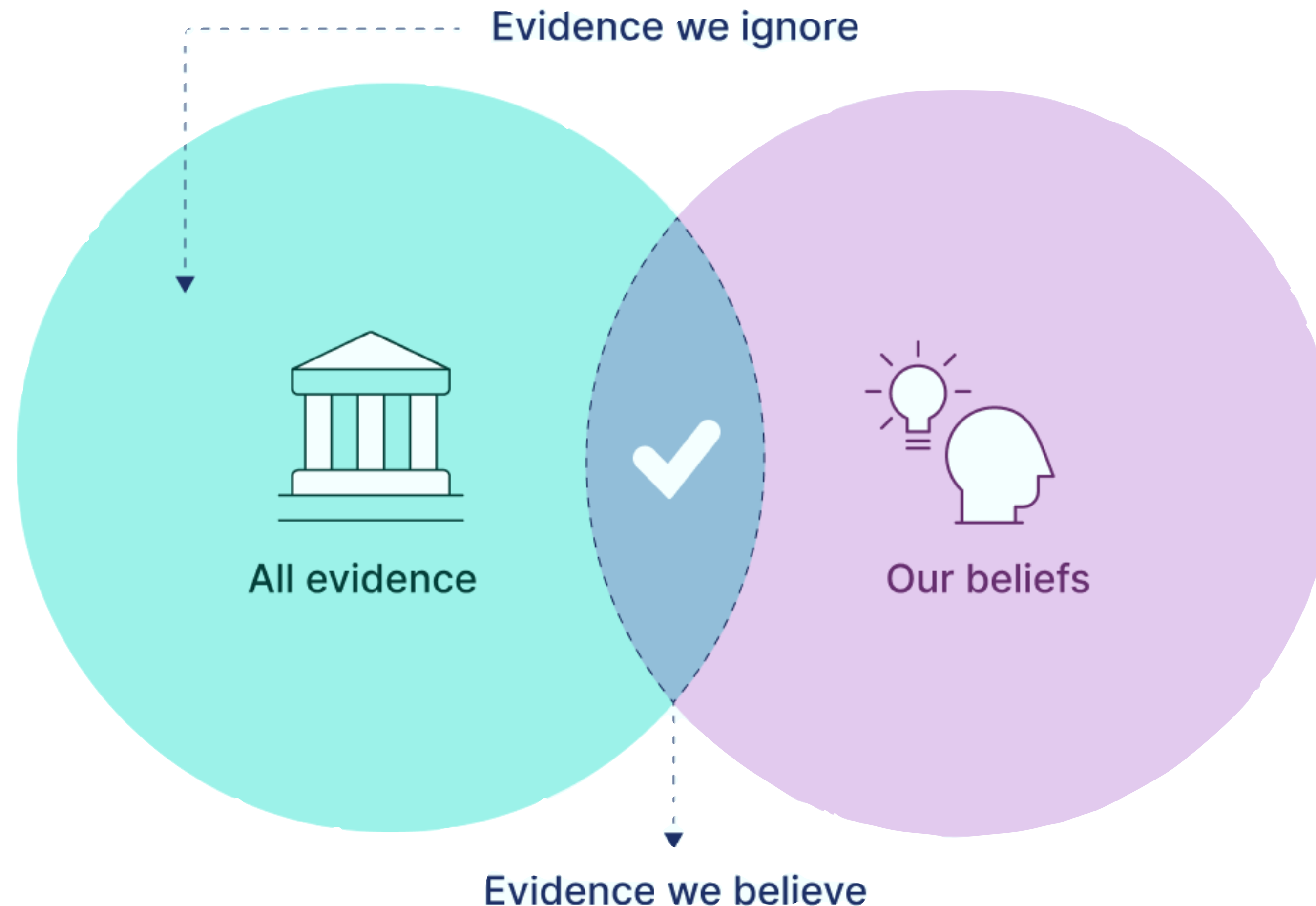
Gestalt grouping

We have **limited** attention and visual perception, think about how much your readers can process

Questions?

Cognitive biases

Confirmation bias



"Confirmation bias is the tendency to search for, interpret, favor, and recall information that confirms or support one's prior personal **beliefs** or values.

Confirmation bias

“Are dogs better than cats?” into an online search engine, articles that argue in favor of dogs will appear first.

Reverse the question and type “are cats better than dogs?”, you will get results in support of cats.

In many people's opinion, dogs are better than cats. Why? Dogs are more trainable, use the bathroom outside, can protect you, and promote a more active lifestyle. Sure, it is possible to think one is better than the other, but still love both. May 6, 2024



 The Spruce Pets <https://www.thesprucepets.com › Dogs › Getting Started>

[10 Reasons Dogs Are Better Than Cats - The Spruce Pets](#)

Compared to dogs, cats are a low maintenance pet. They don't require formal training, they don't need to be taken out multiple times a day, and they're even able to take care of basic self-cleaning.

 Freshpet <https://www.freshpet.com › blog › 10-reasons-why-cats-...>

[10 Reasons Why Cats Make Great Pets - Freshpet](#)

Anchoring bias

Where individuals rely too heavily on the first piece of information (the "anchor") when making decisions, judgments, or estimates.



Priming effects

Our exposure to a certain stimulus influences their response to a **subsequent** prompt, without any awareness of the connection.

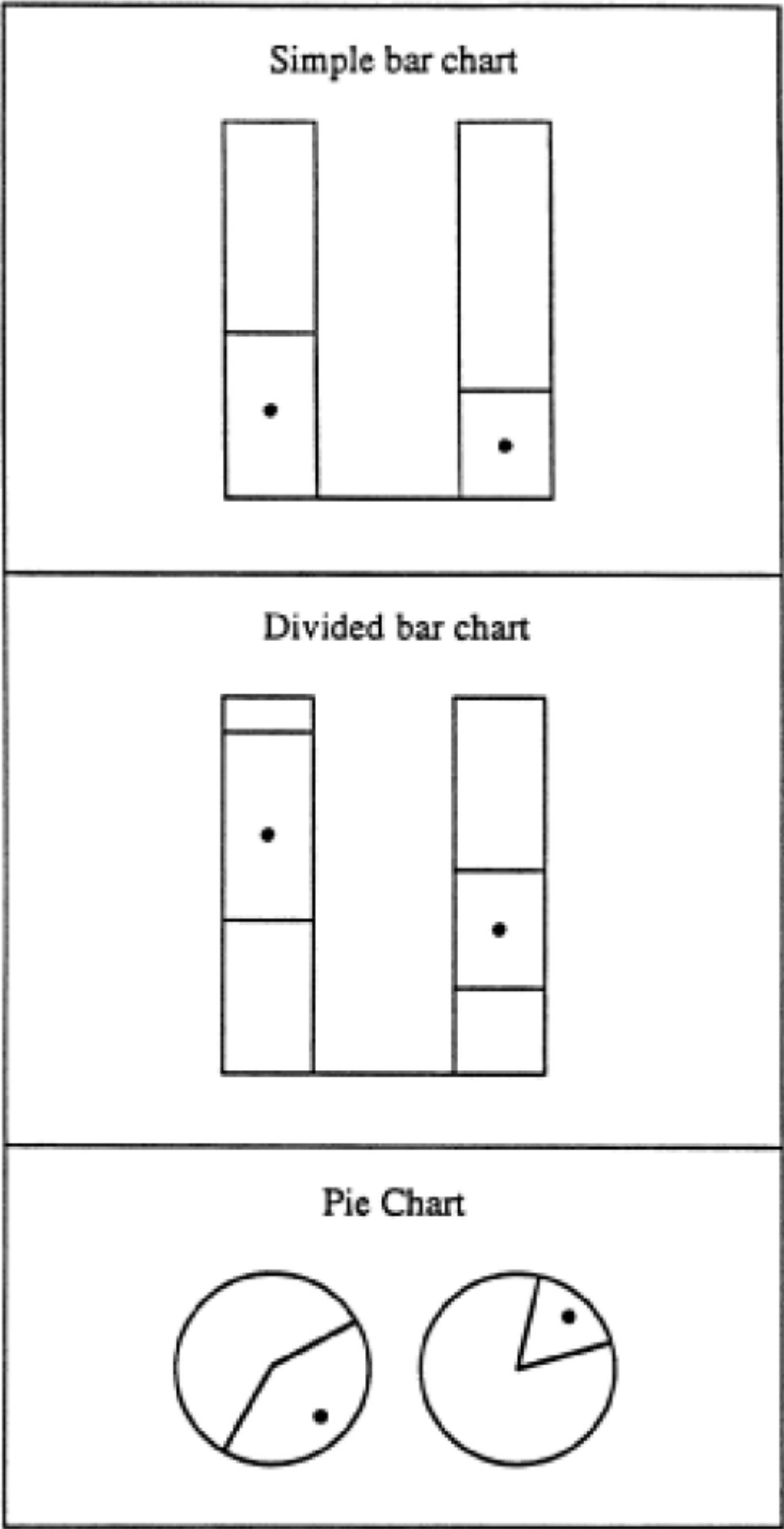
These stimuli are often related to **words** or **images** that people see during their day-to-day lives.

Just watched a documentary about pizza



Me: maybe I should have pizza as dinner...

Priming effects



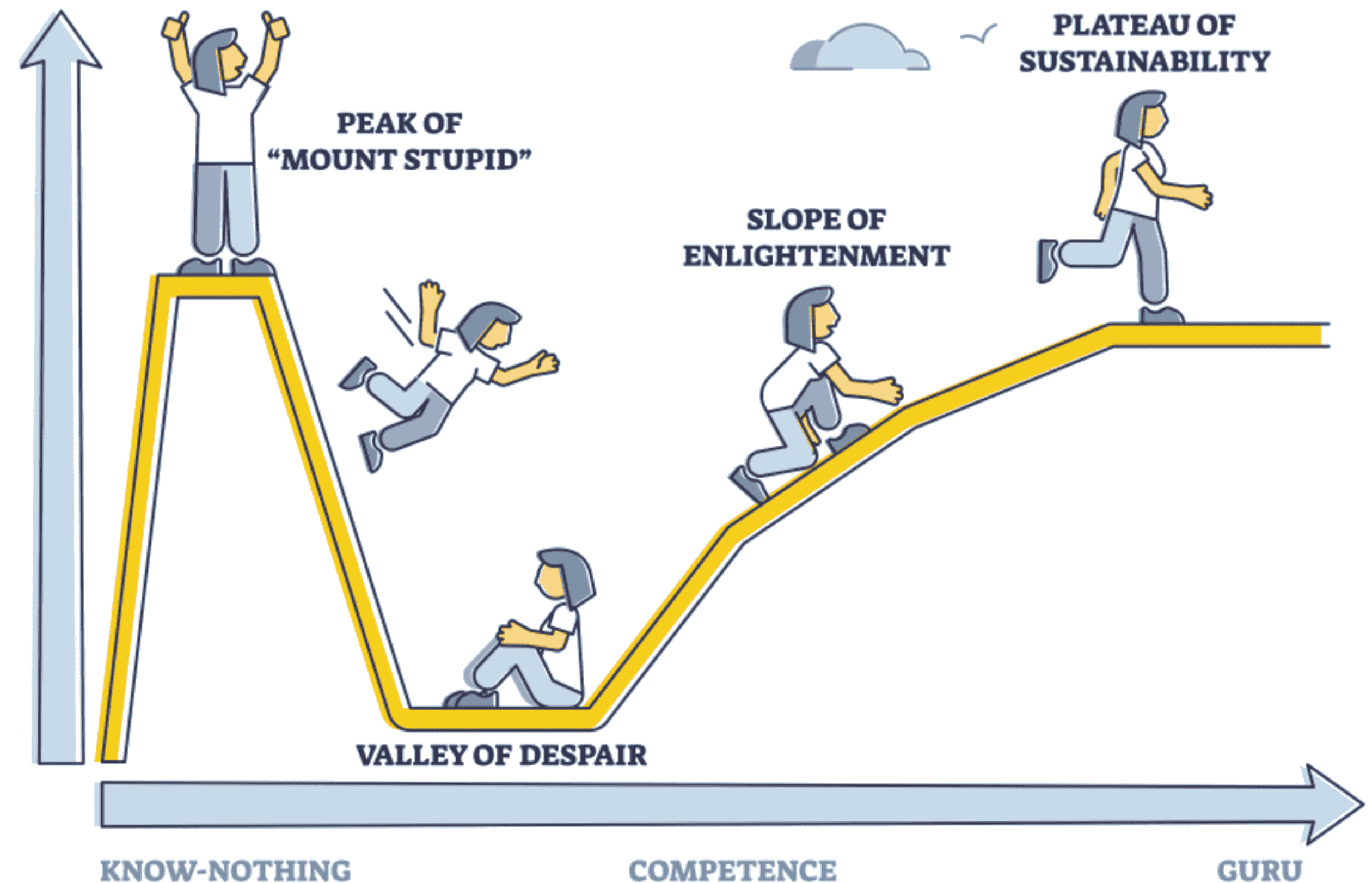
better

worse

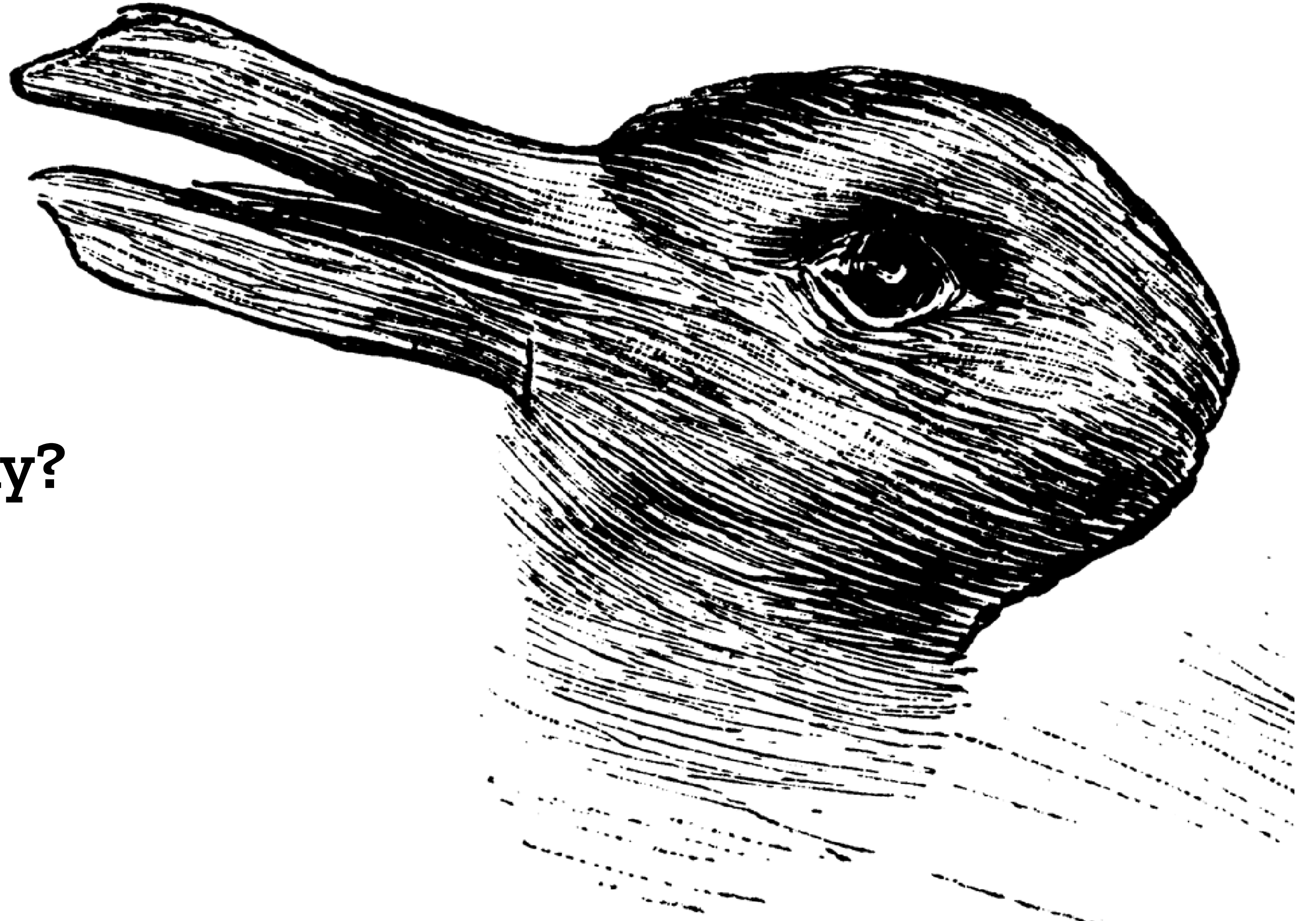
Dunning–Kruger effect

People with lower ability tend to overestimate their competence

People with higher ability tend to underestimate their relative performance



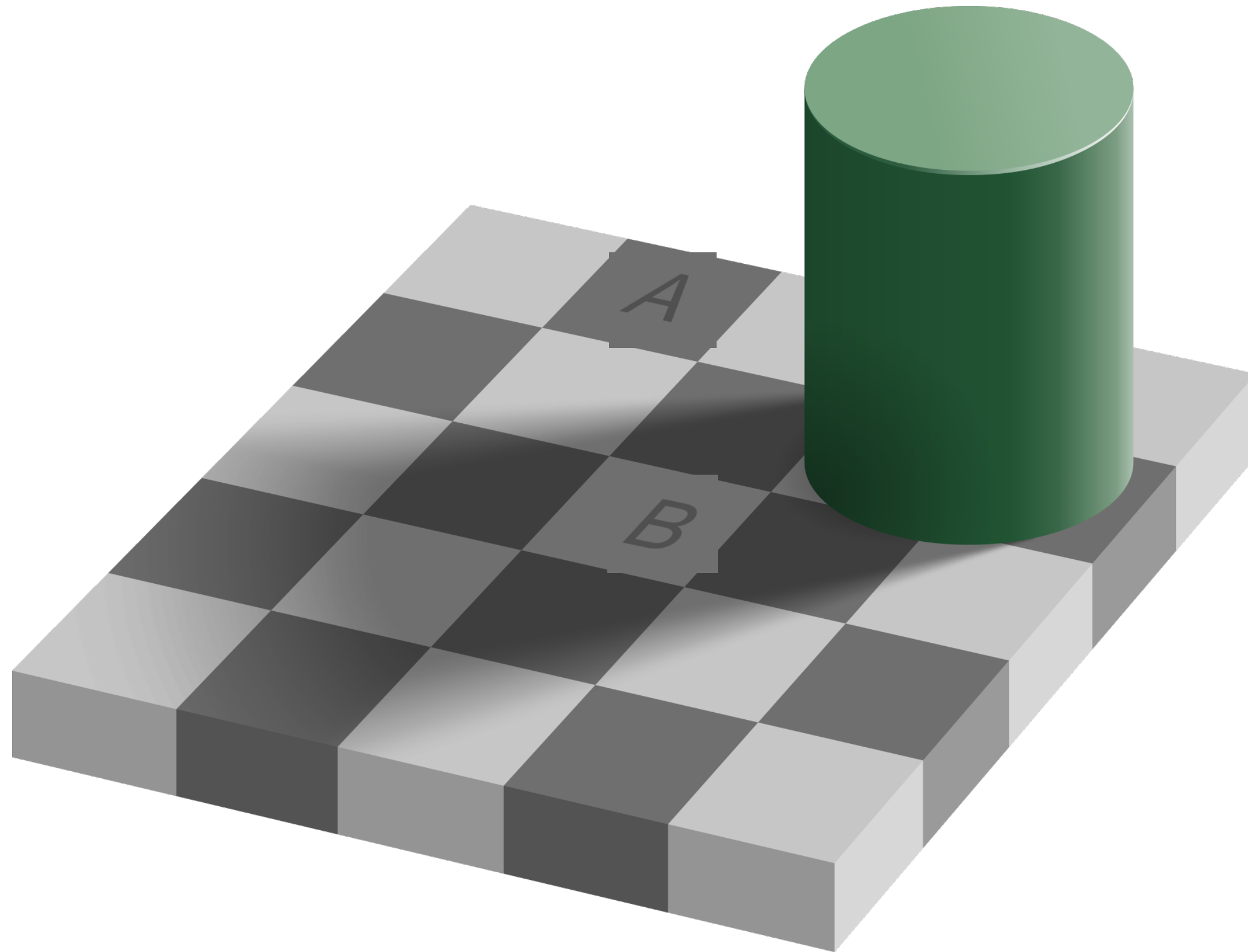
Visual illusion



Duck or Bunny?

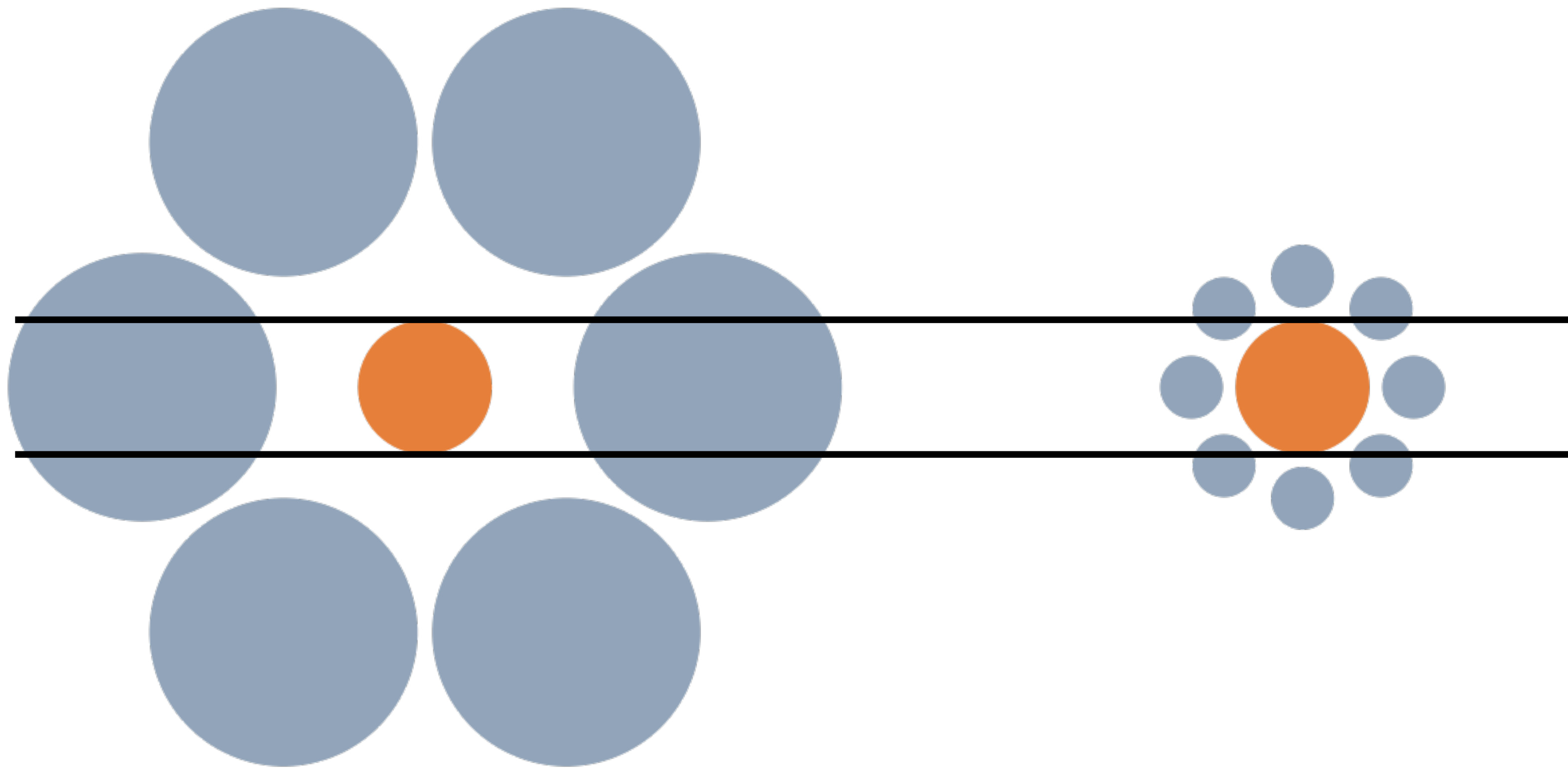
Adelson's checker shadow illusion

A is darker than B?



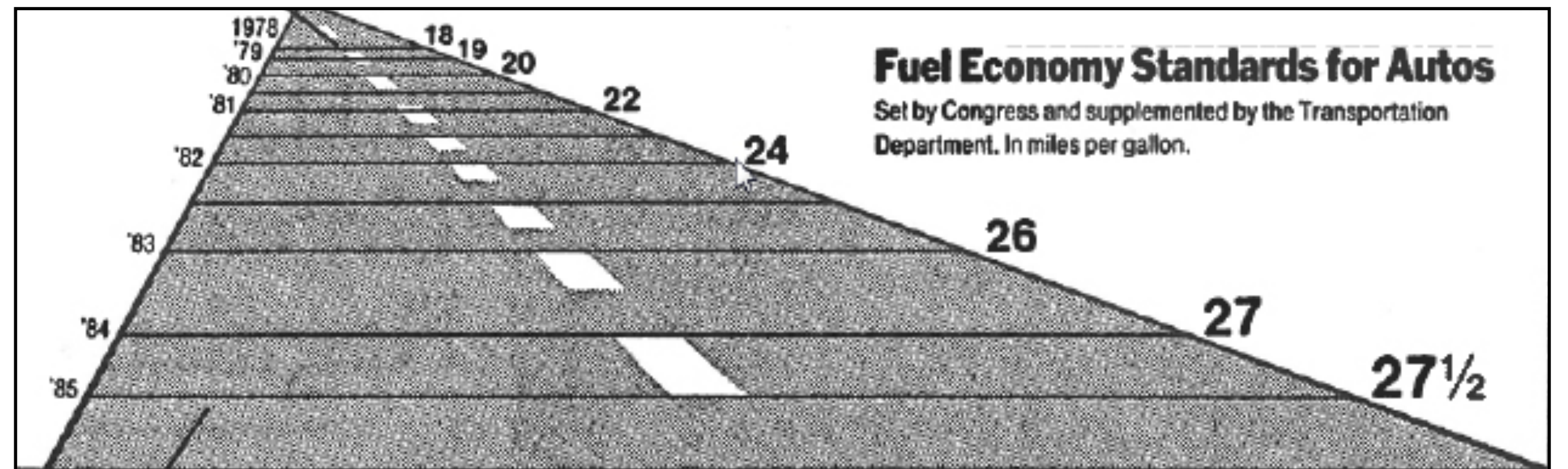
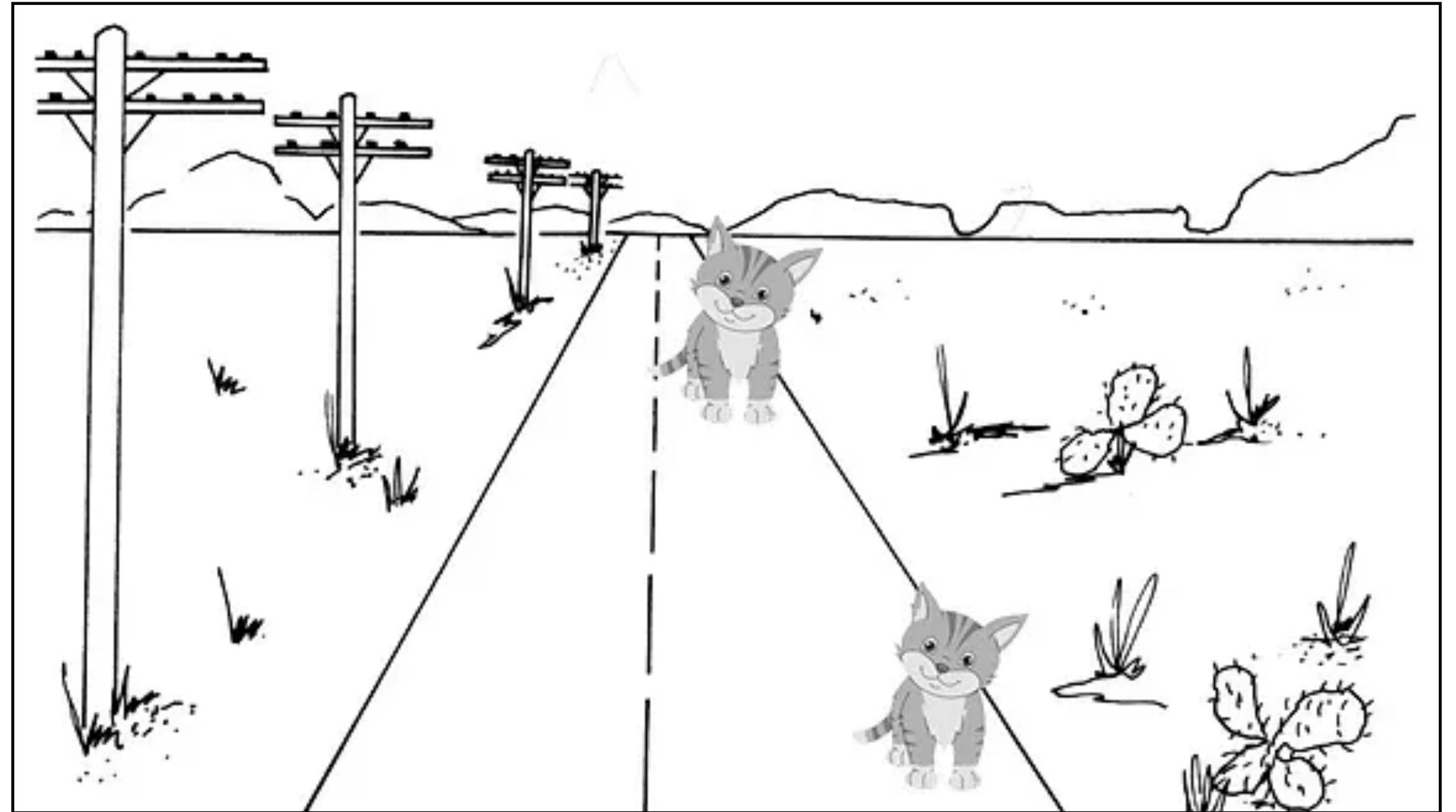
Ebbinghaus illusion

Left **orange** circle is bigger?



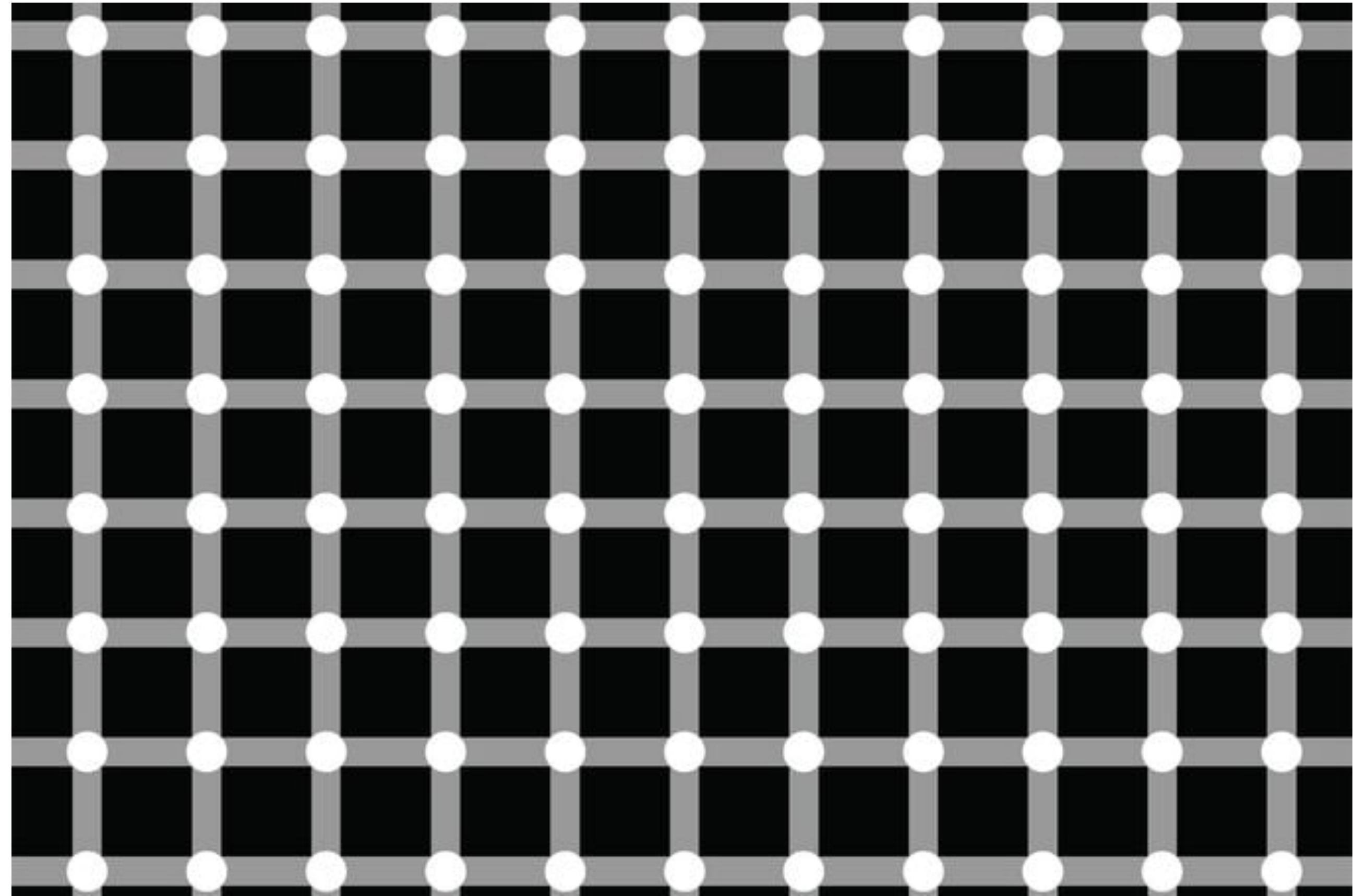
Ponzo illusion

Things look **smaller** when they are **further away**, so our brain **compensates** for this, by making things that we know are far away seem bigger.



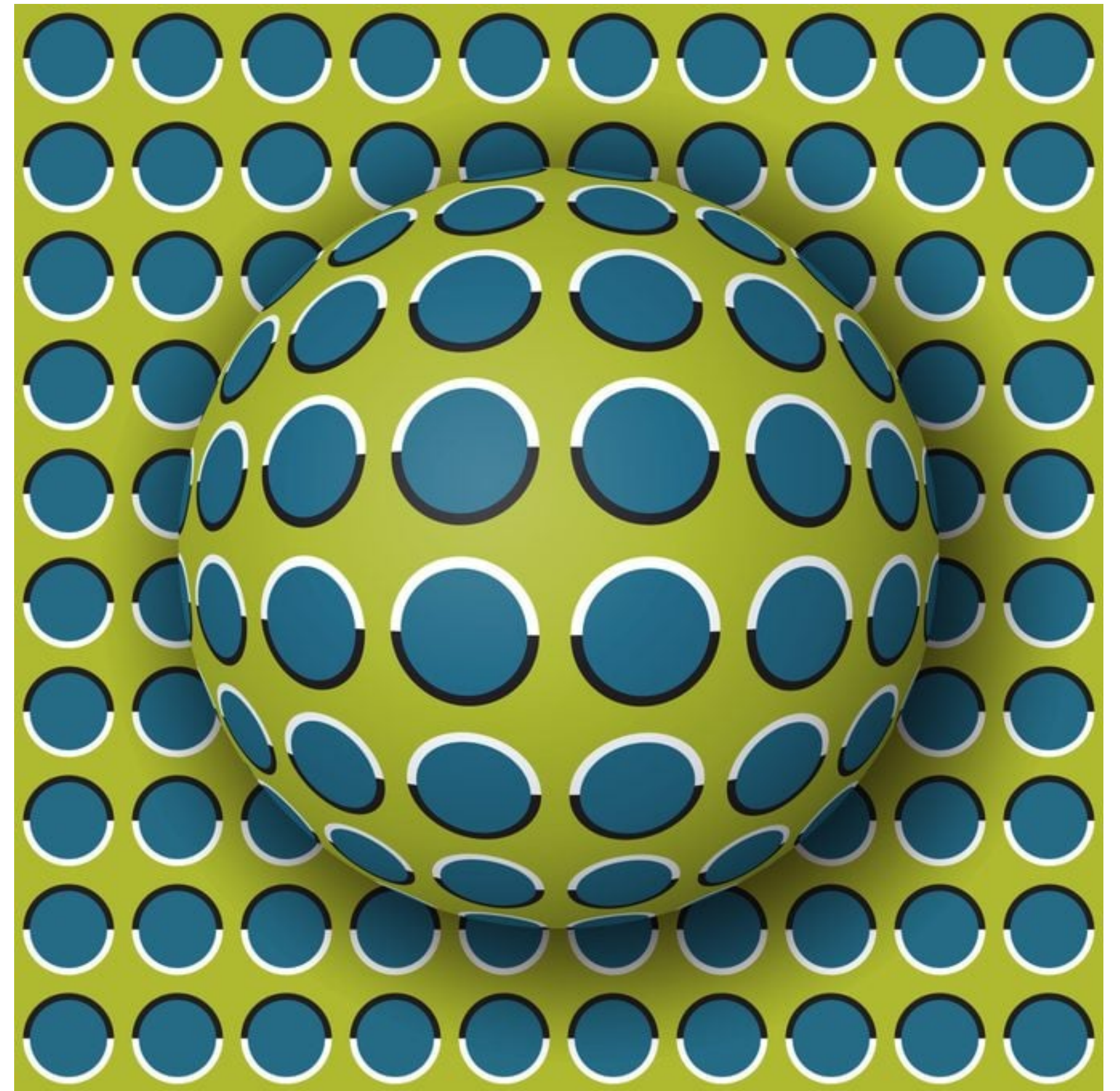
Scintillating grid illusion

A neural process called lateral inhibition. The intensity at a point in the visual system is not simply the result of a single receptor, but the result of a group of receptors.



Moving right along

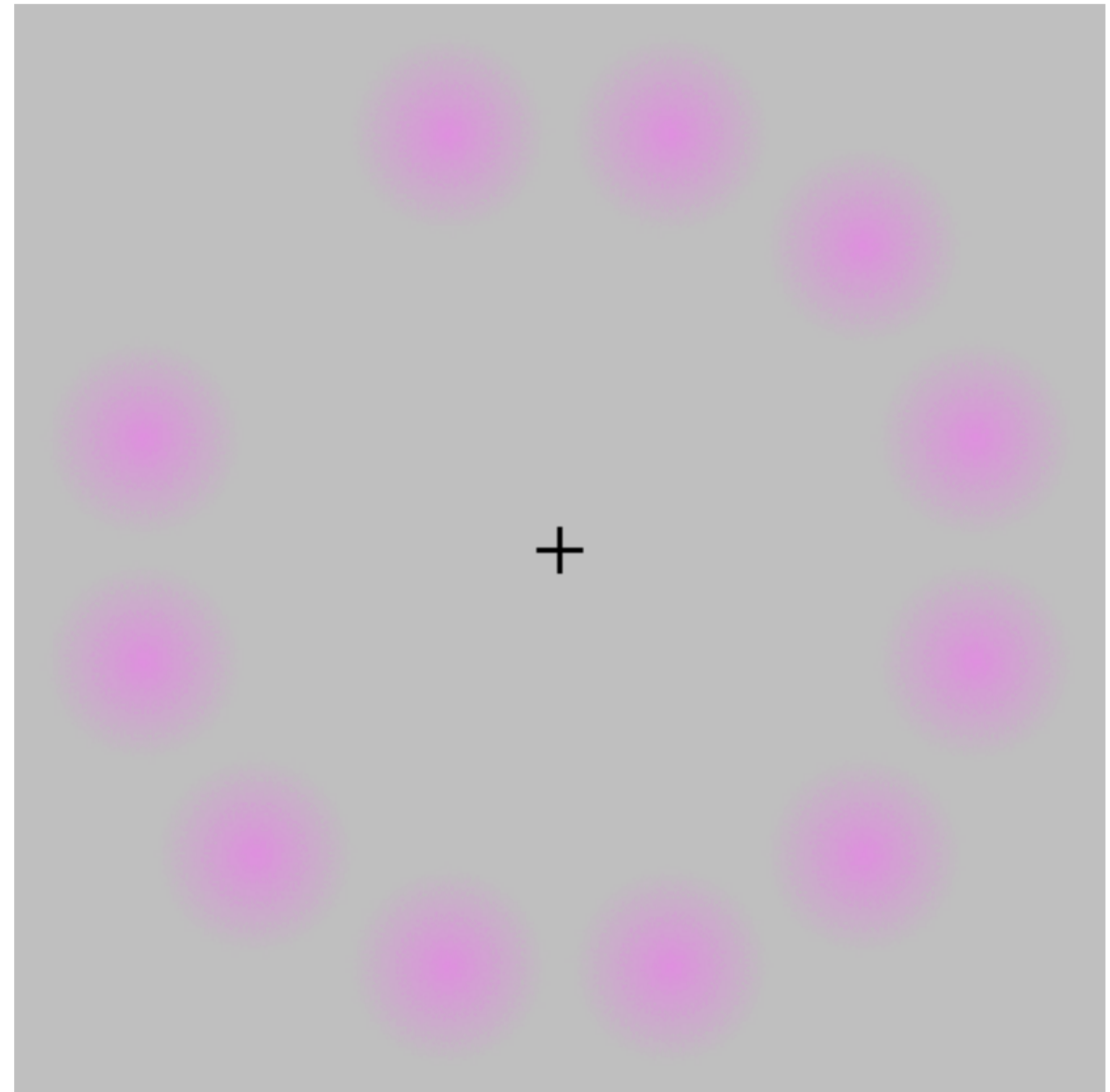
The sphere is going downward as if it were on a conveyor belt or a slow treadmill....?



Vanishing colors (Troxler effect)

Stare at the cross in the centre of this gif, and keep your eyes as still as possible- what happens?

A stationary stimulus in your peripheral vision fades away and disappears when you fixate on a central point for a period of time.



Recap

Signal detection

Magnitude estimation

Pre-attentive processing

Selective attention

Change blindness

Gestalt grouping

Web lab

Lab 3: Understanding the General Update Pattern

Add point

Remove point

Update points

